

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

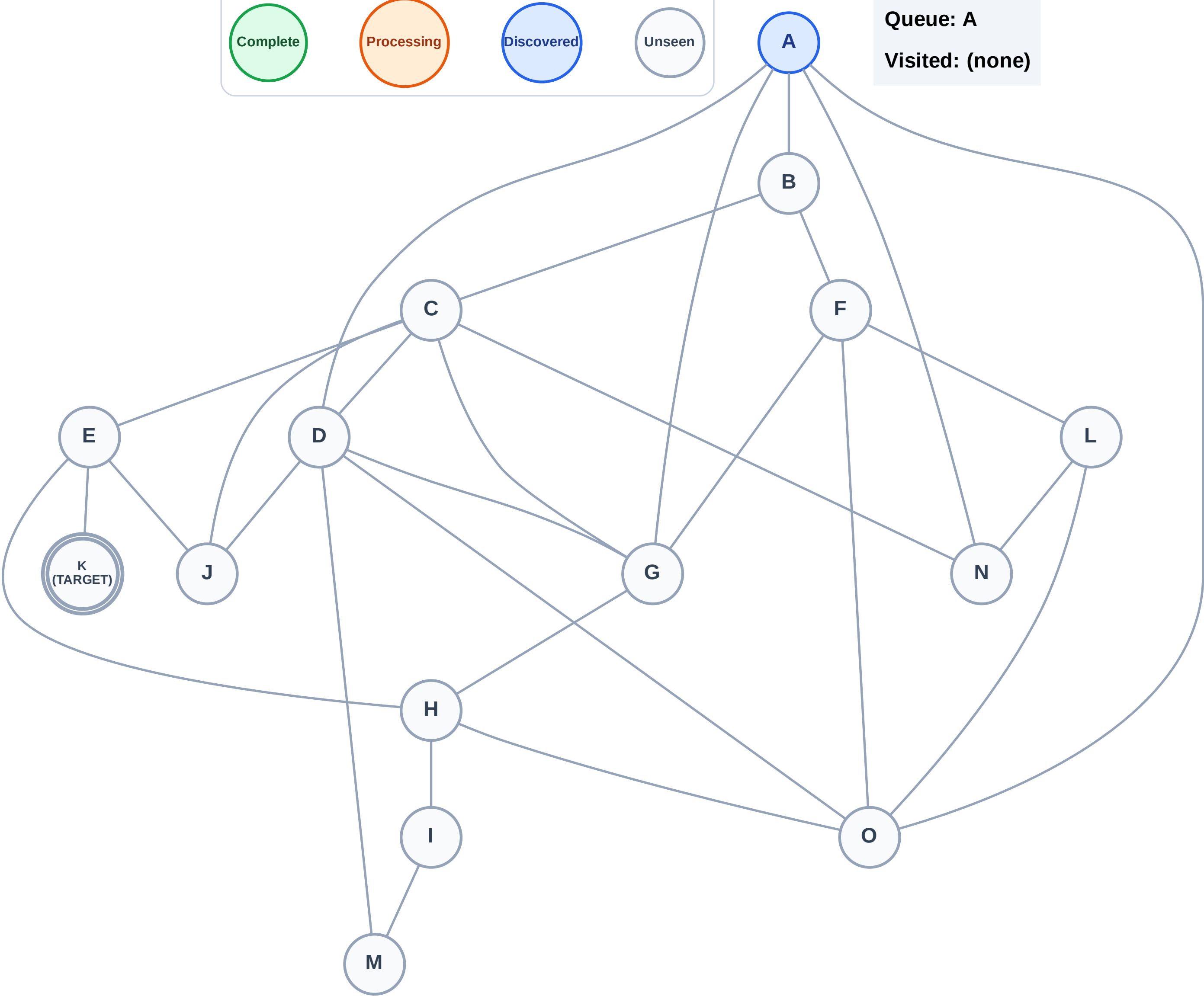
Complete

Processing

Discovered

Unseen

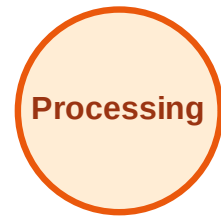
Queue: A
Visited: (none)



BFS Traversal

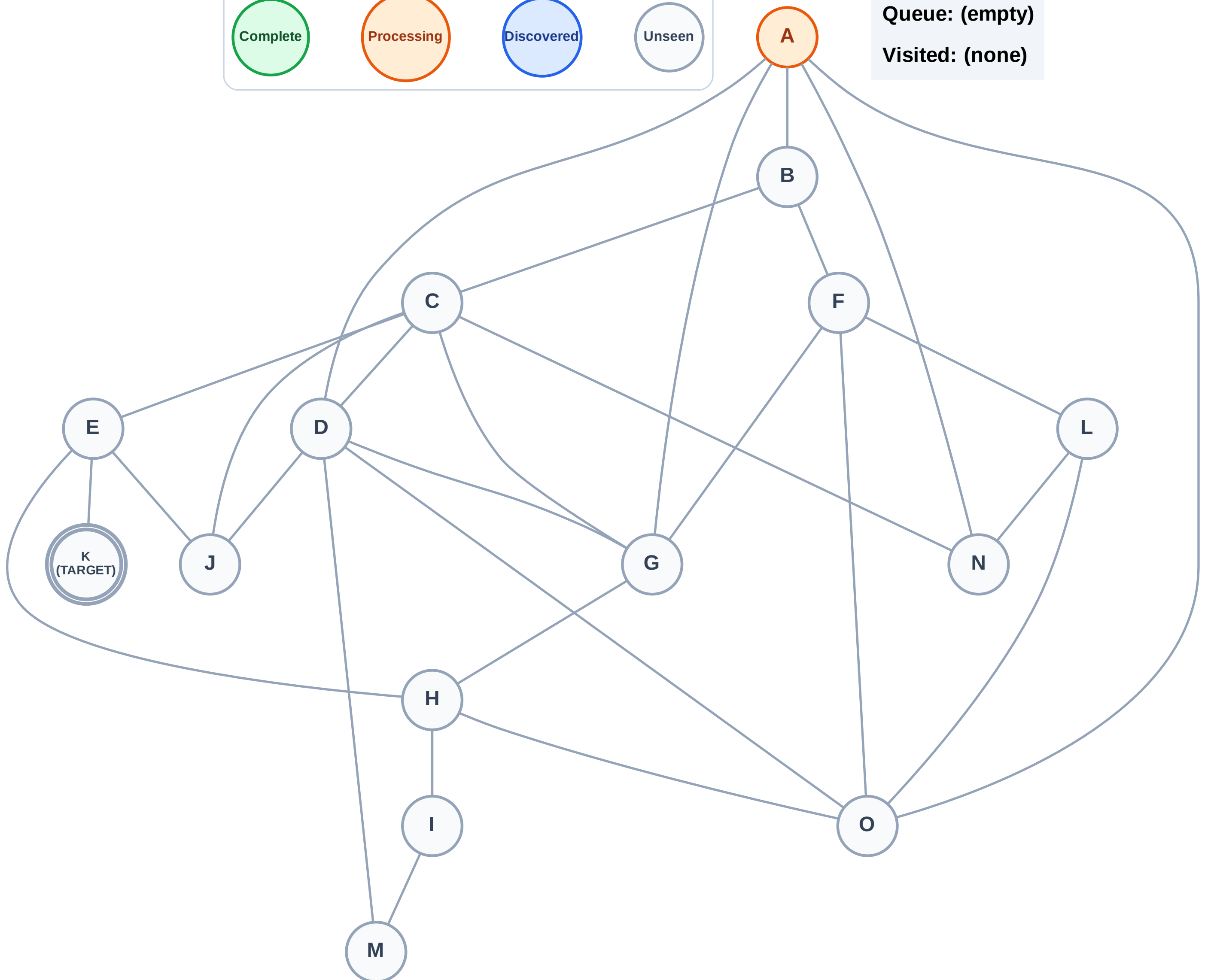
Step 2: Process node A

Legend



Queue: (empty)

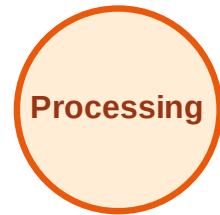
Visited: (none)



BFS Traversal

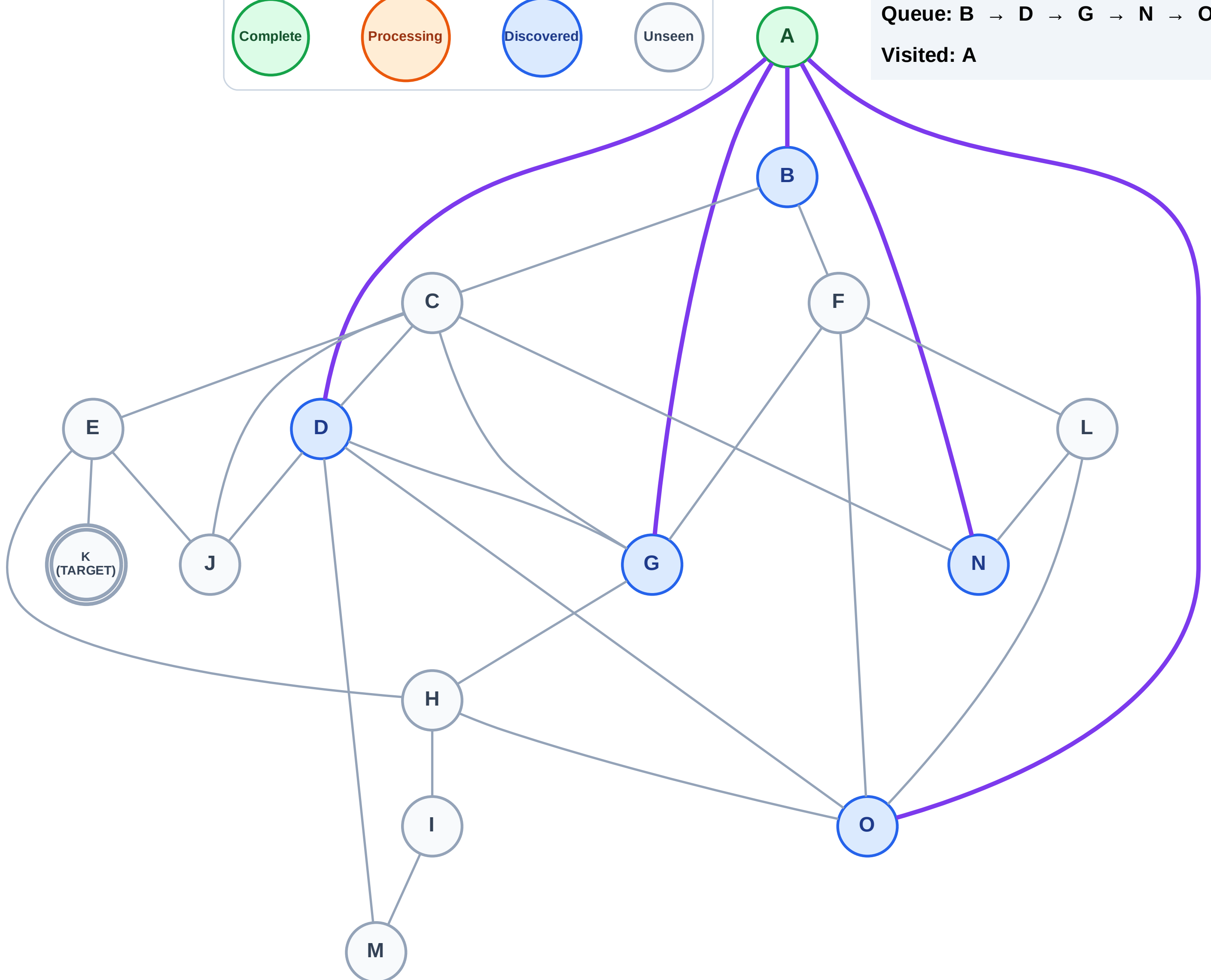
Step 3: Discover B, D, G, N, O from A

Legend



Queue: B → D → G → N → O

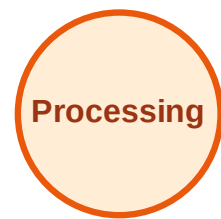
Visited: A



BFS Traversal

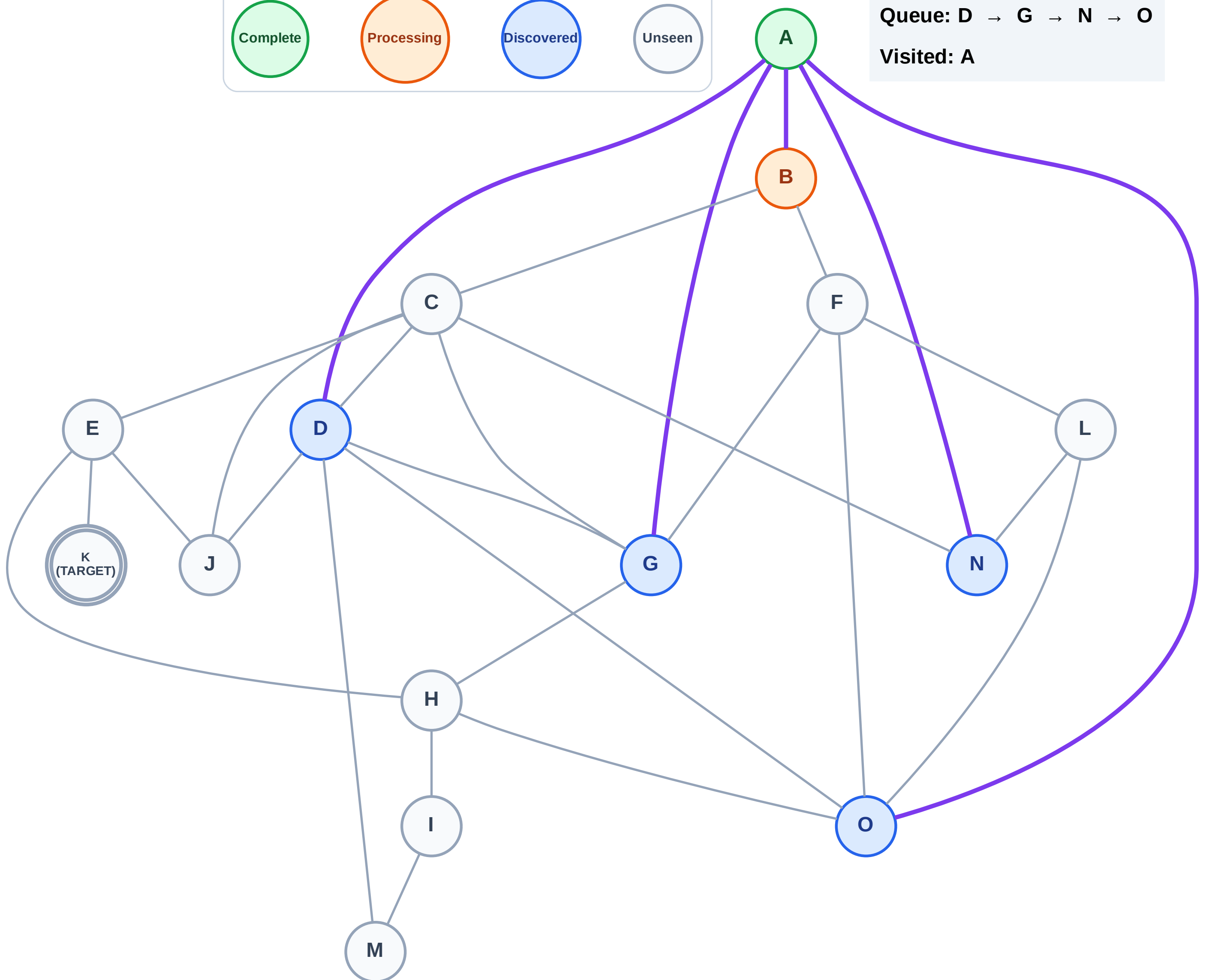
Step 4: Process node B

Legend



Queue: D → G → N → O

Visited: A



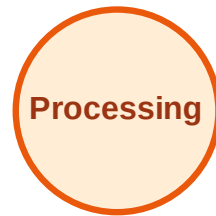
BFS Traversal

Step 5: Discover C, F from B

Legend



Complete



Processing



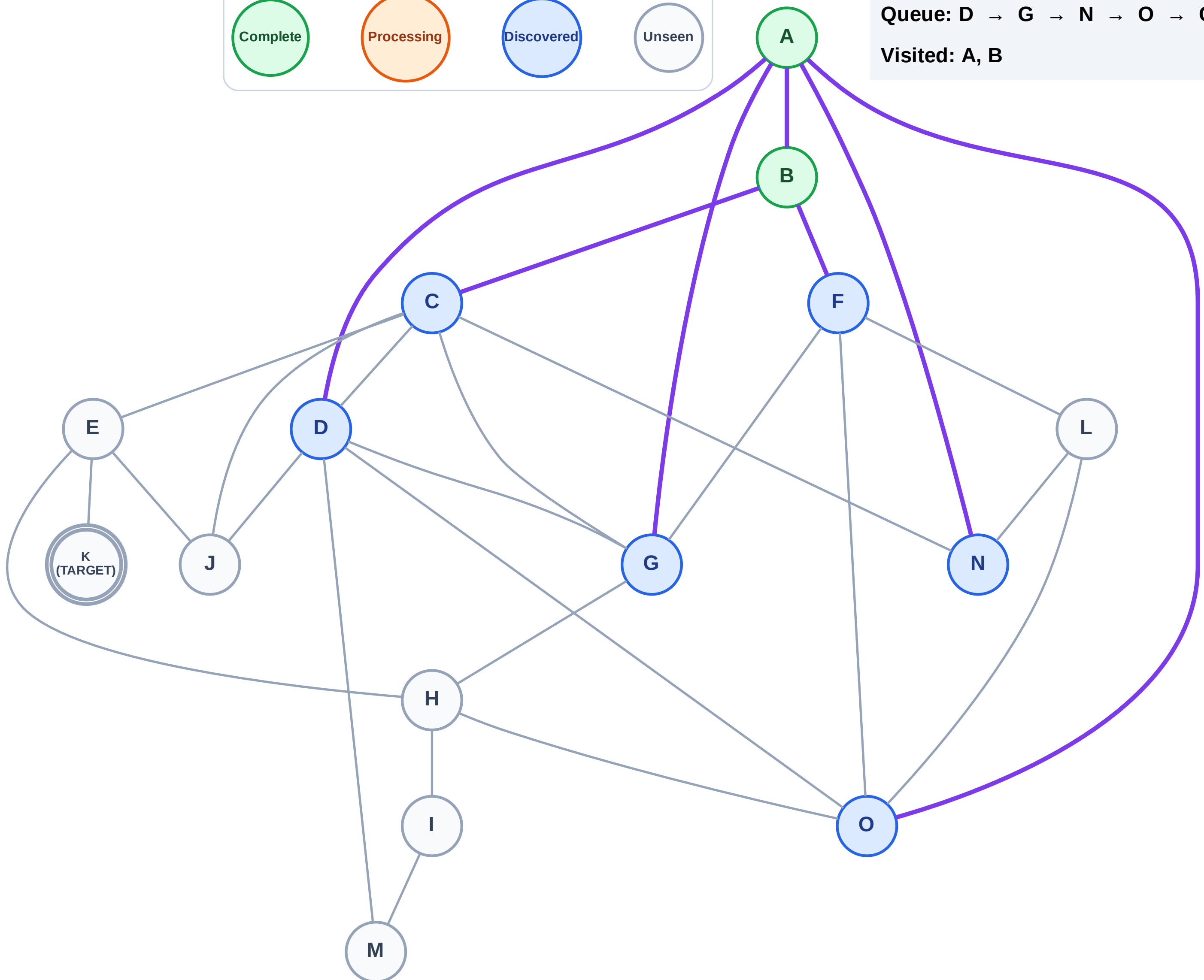
Discovered



Unseen

Queue: D → G → N → O → C → F

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

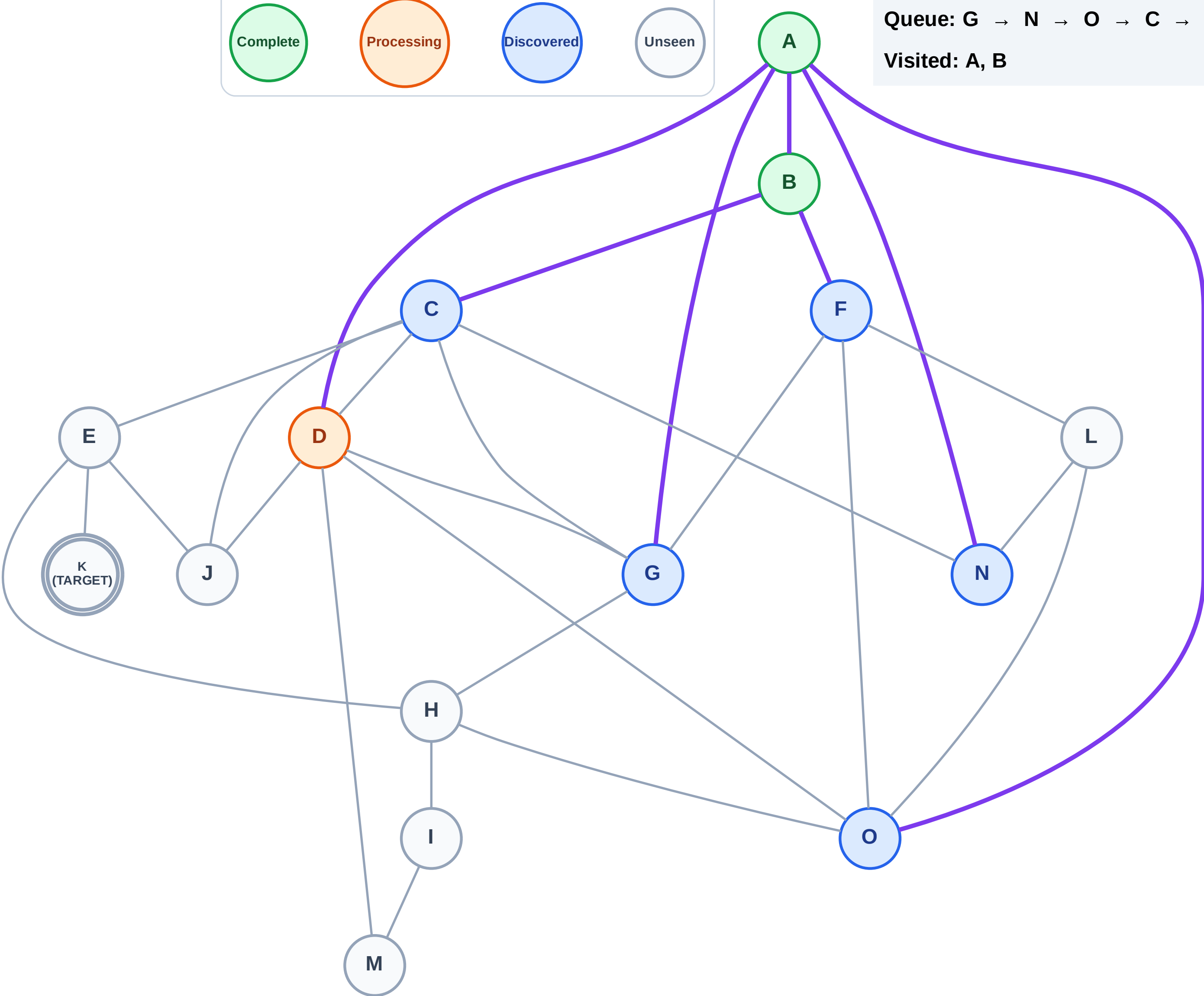
Processing

Discovered

Unseen

Queue: G → N → O → C → F

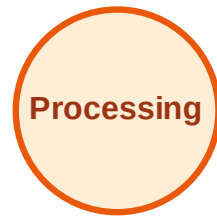
Visited: A, B



BFS Traversal

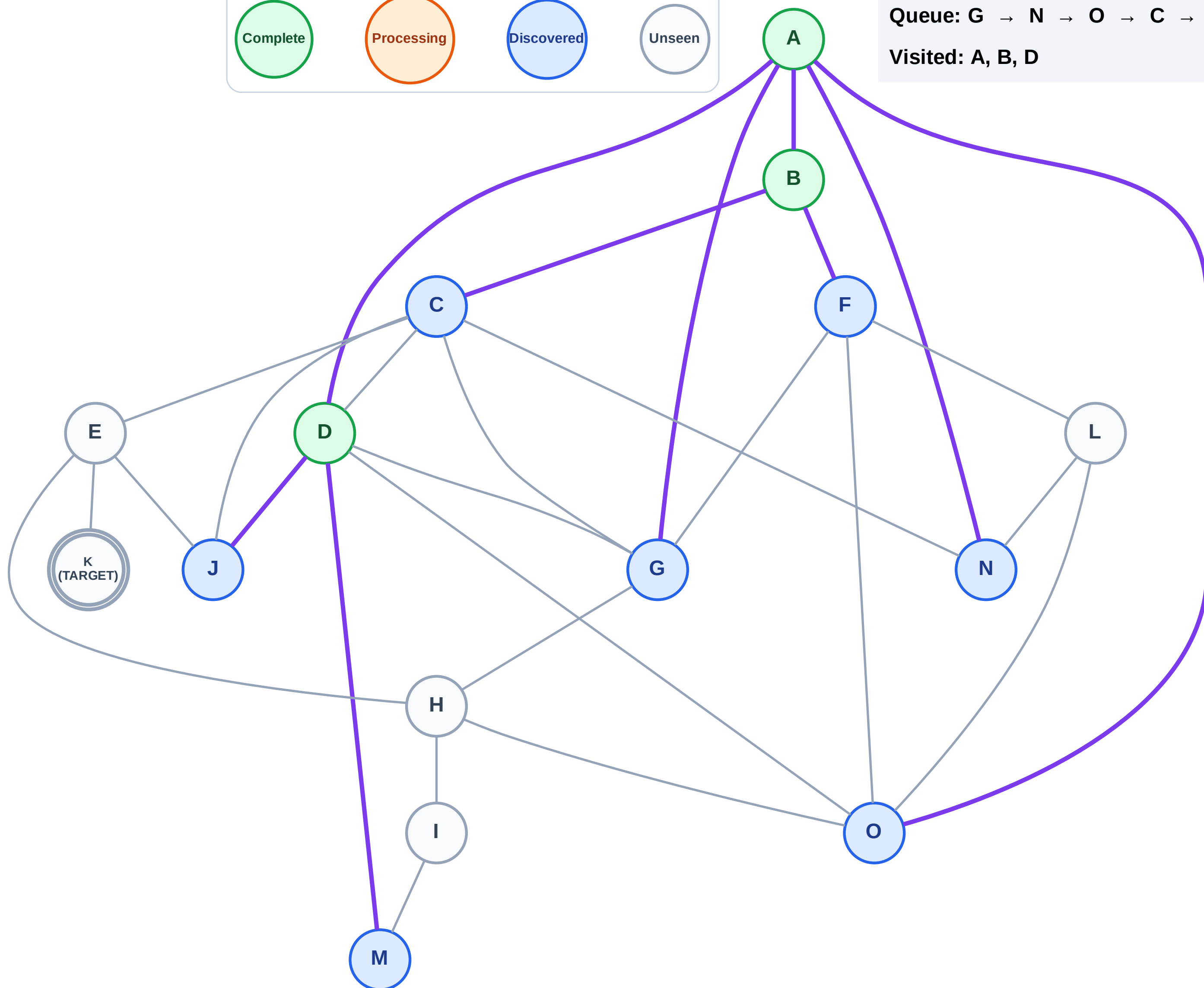
Step 7: Discover J, M from D

Legend



Queue: G → N → O → C → F → J → M

Visited: A, B, D



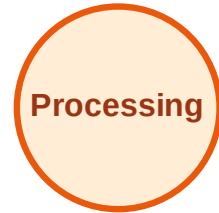
BFS Traversal

Step 8: Process node G

Legend



Complete



Processing



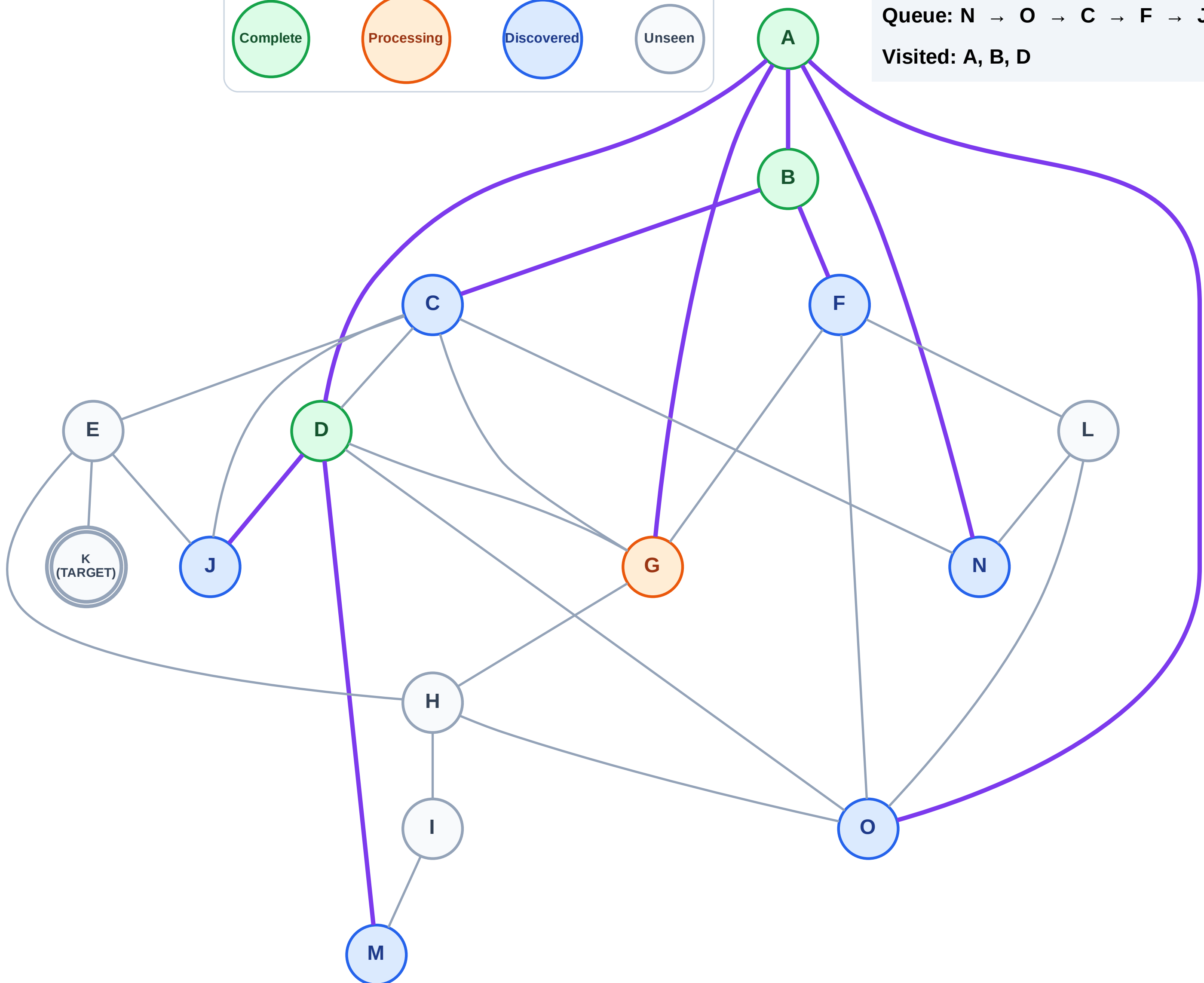
Discovered



Unseen

Queue: N → O → C → F → J → M

Visited: A, B, D



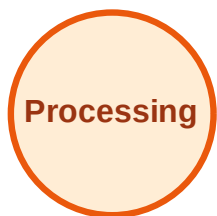
BFS Traversal

Step 9: Discover H from G

Legend



Complete



Processing



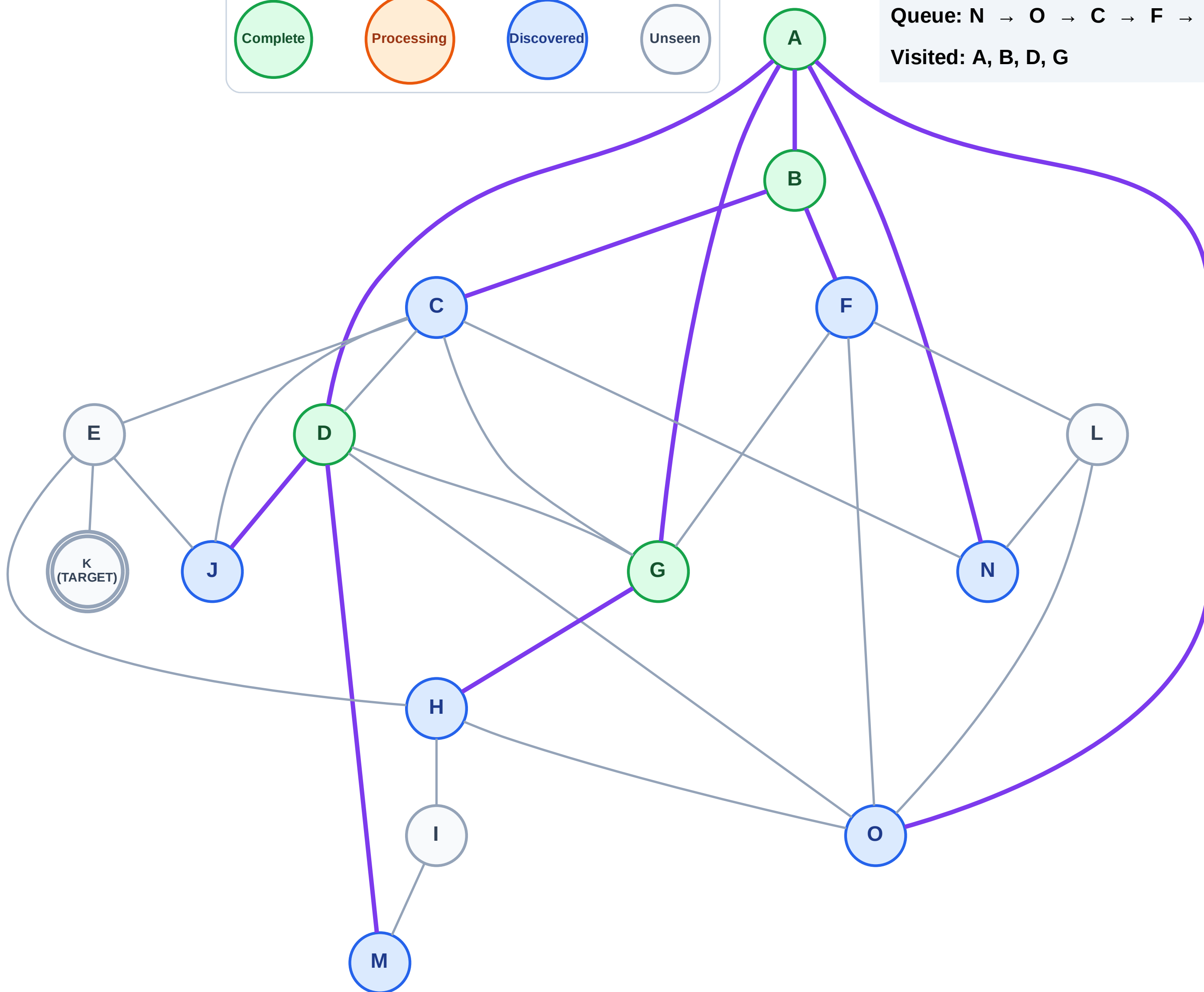
Discovered



Unseen

Queue: N → O → C → F → J → M → H

Visited: A, B, D, G



BFS Traversal

Step 10: Process node N

Legend

Complete

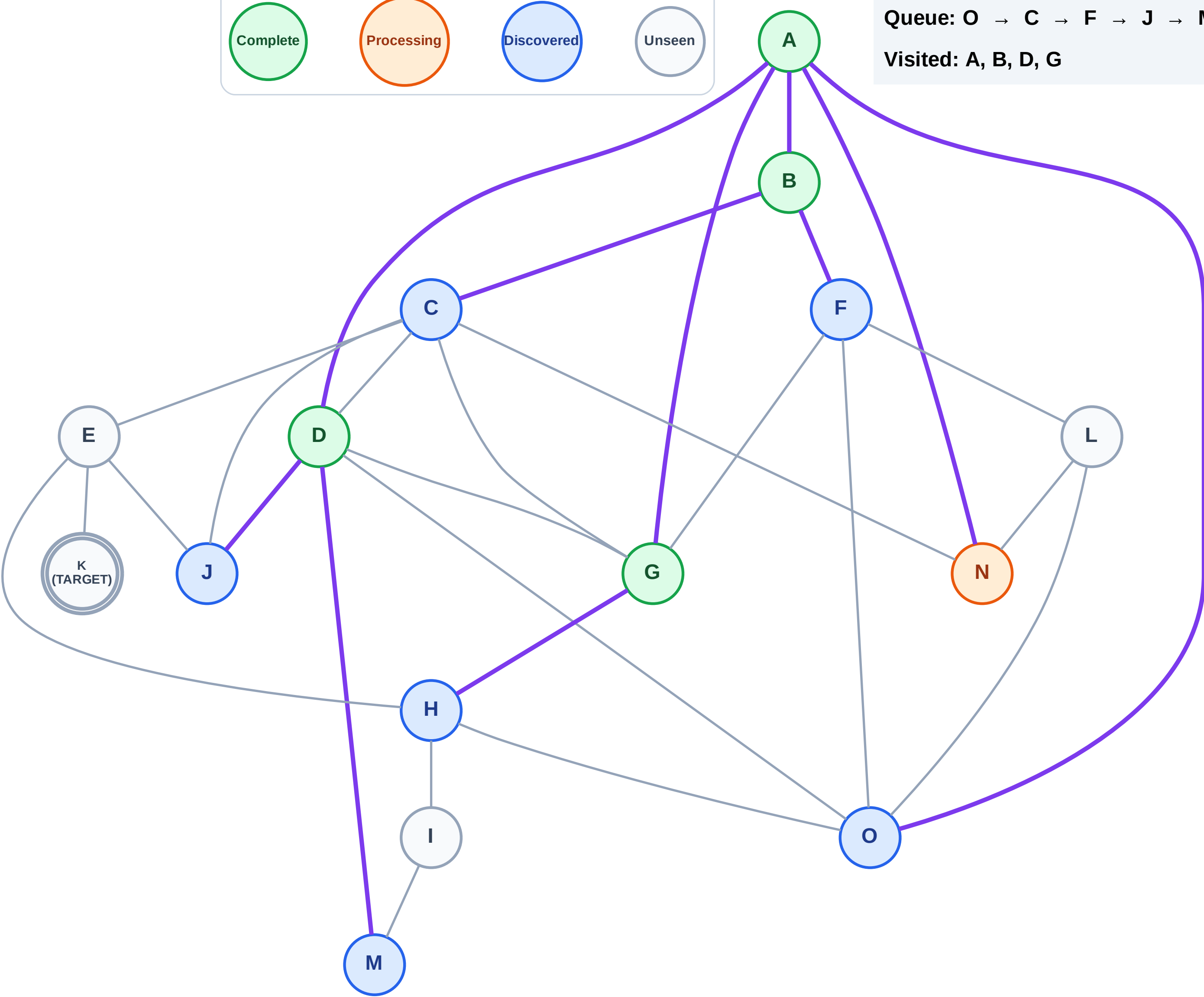
Processing

Discovered

Unseen

Queue: O → C → F → J → M → H

Visited: A, B, D, G



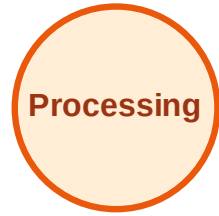
BFS Traversal

Step 11: Discover L from N

Legend



Complete



Processing



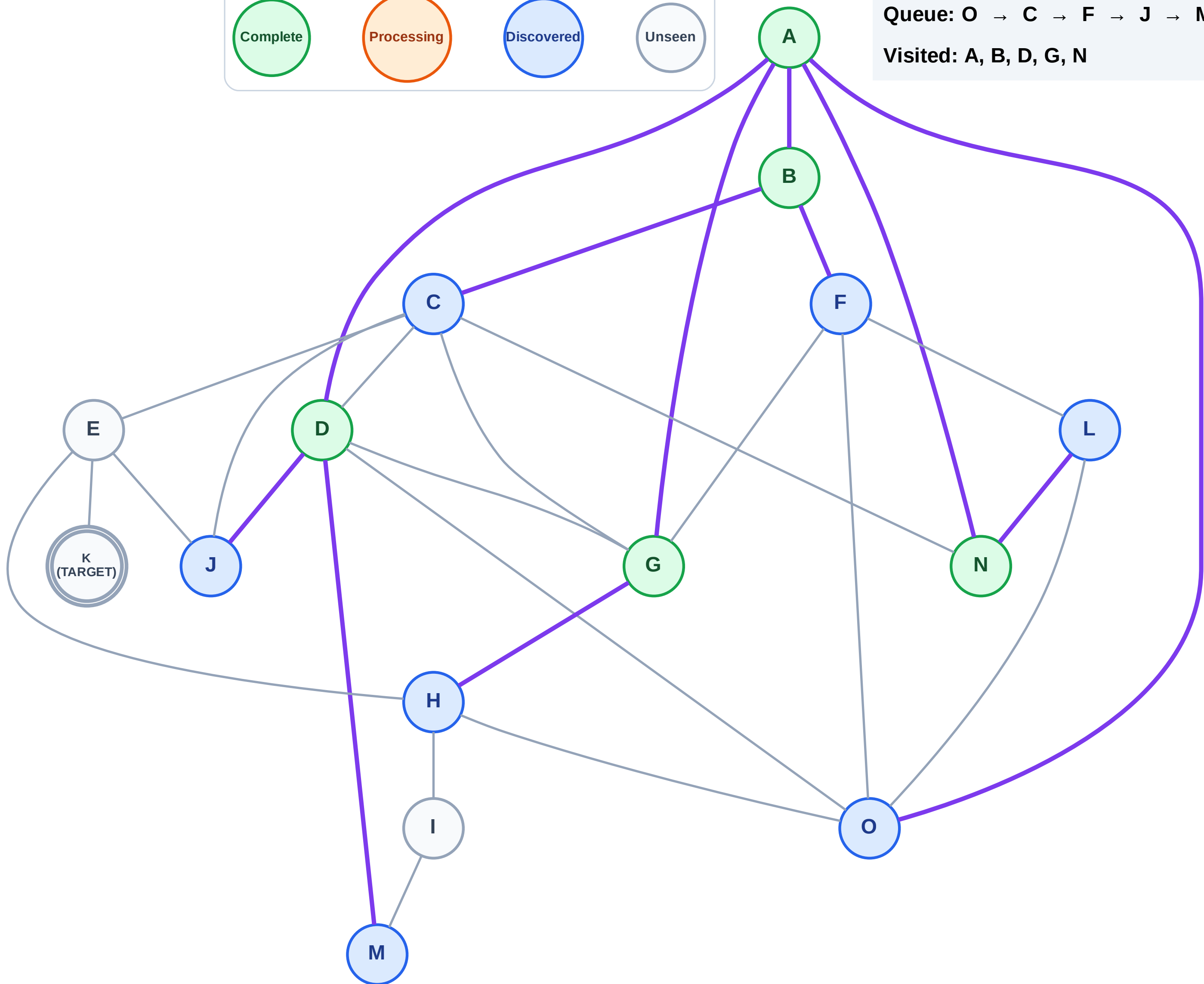
Discovered



Unseen

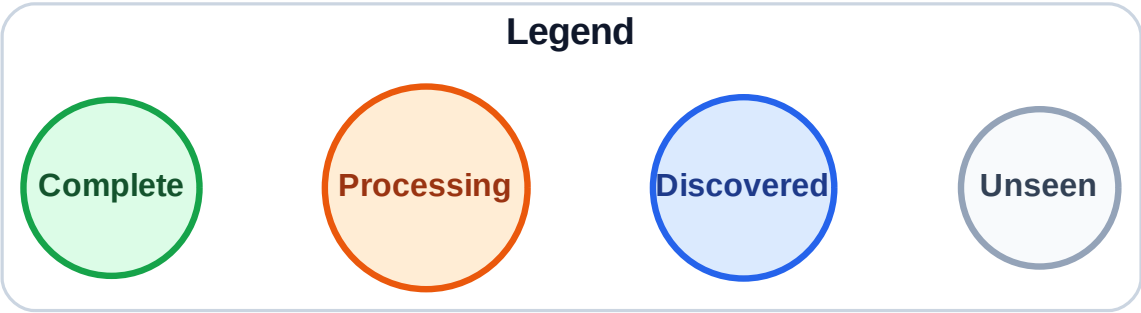
Queue: O → C → F → J → M → H → L

Visited: A, B, D, G, N

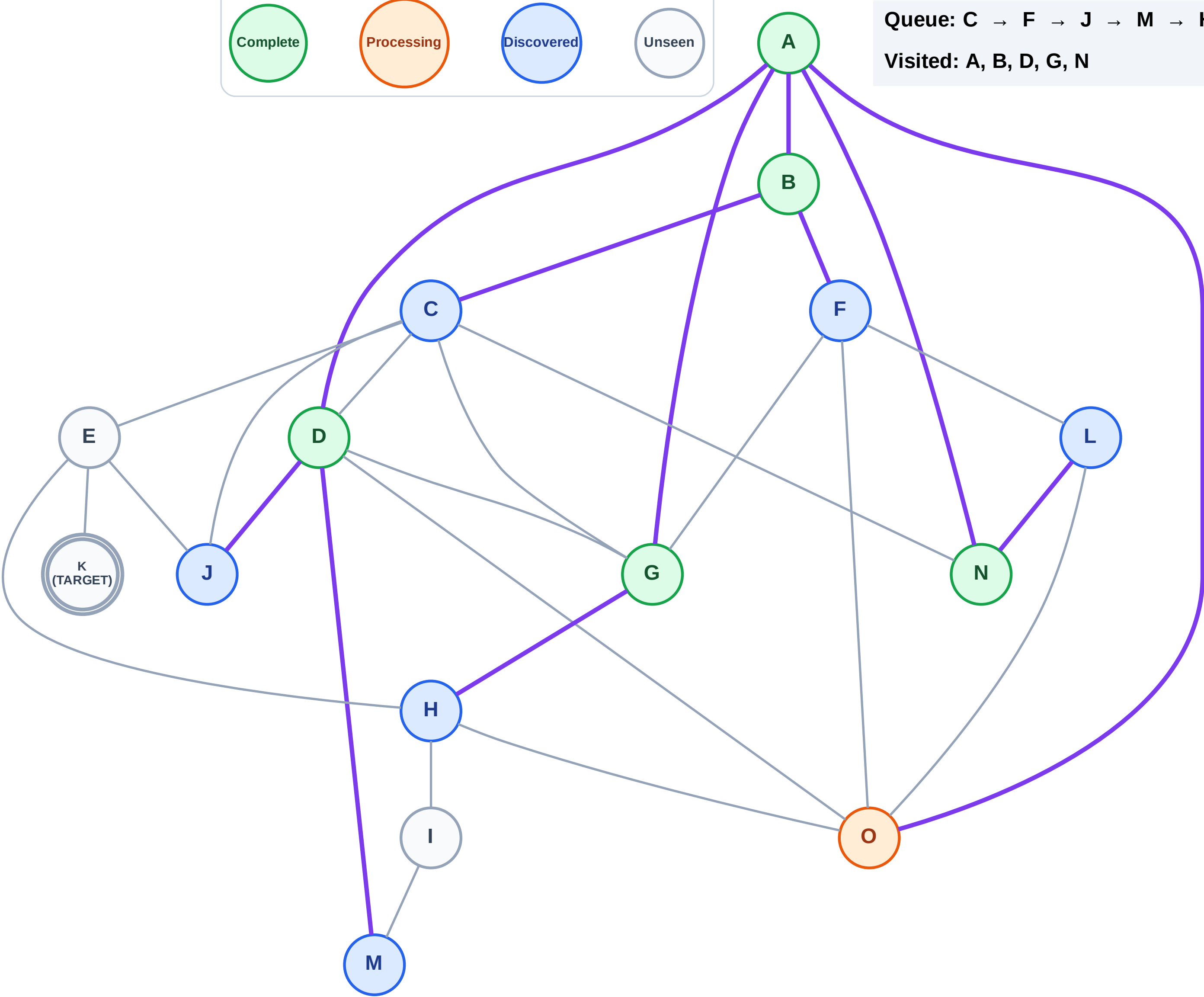


BFS Traversal

Step 12: Process node O



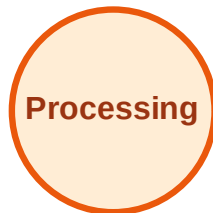
Queue: C → F → J → M → H → L
Visited: A, B, D, G, N



BFS Traversal

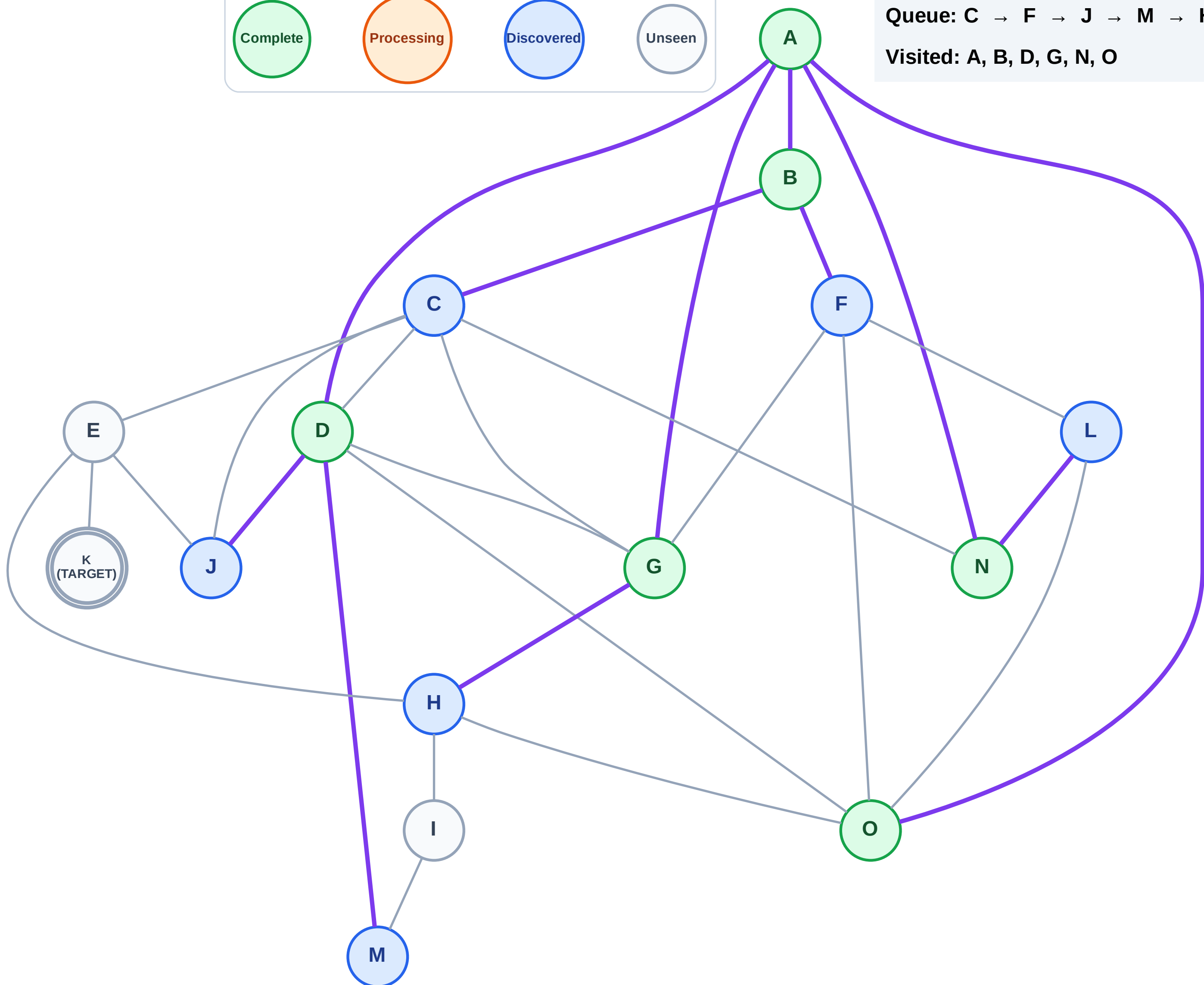
Step 13: No new neighbors from O

Legend



Queue: C → F → J → M → H → L

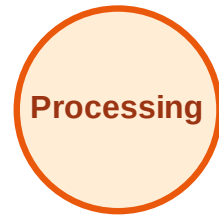
Visited: A, B, D, G, N, O



BFS Traversal

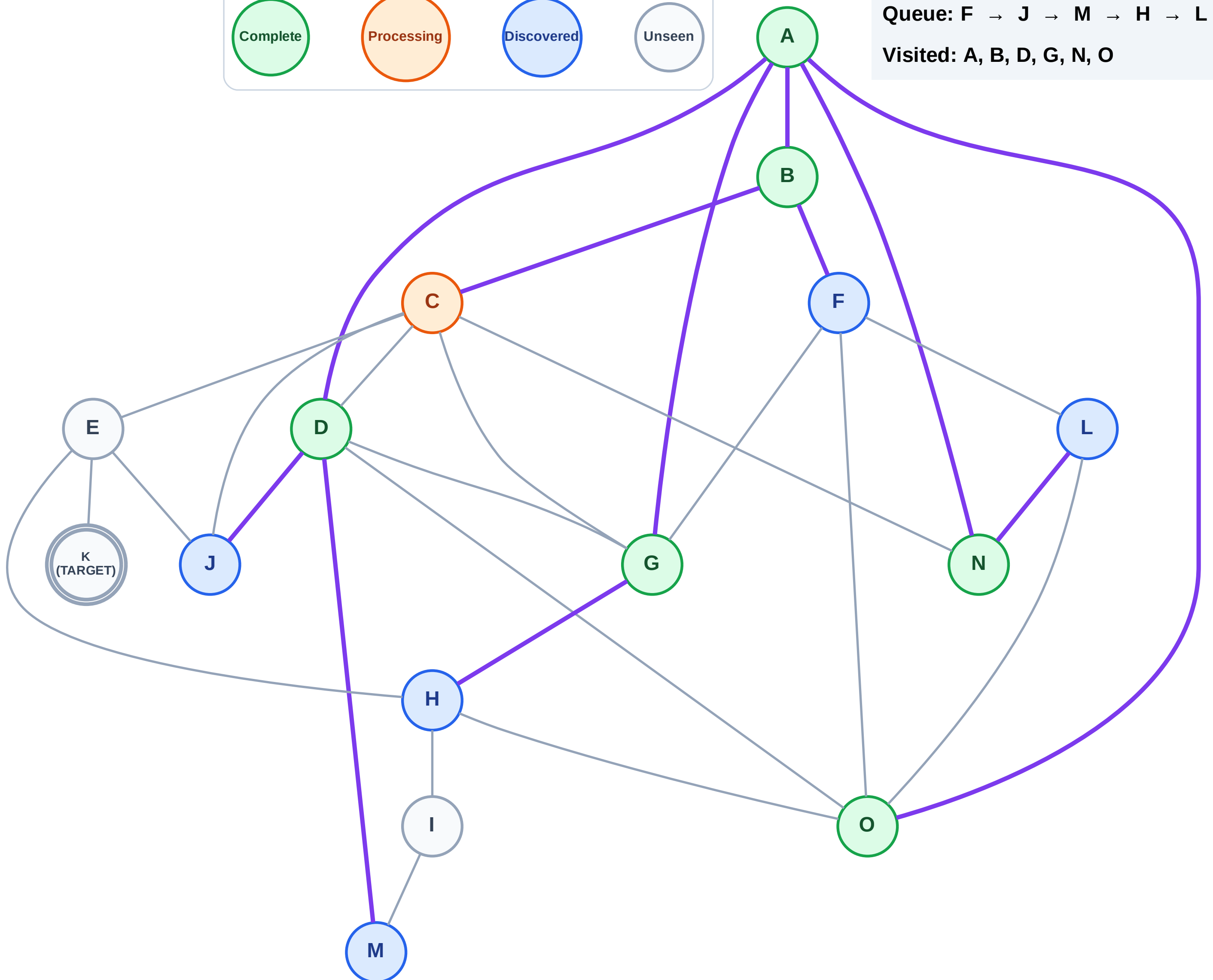
Step 14: Process node C

Legend



Queue: F → J → M → H → L

Visited: A, B, D, G, N, O



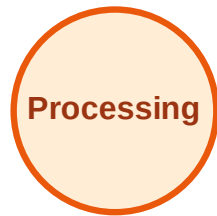
BFS Traversal

Step 15: Discover E from C

Legend



Complete



Processing



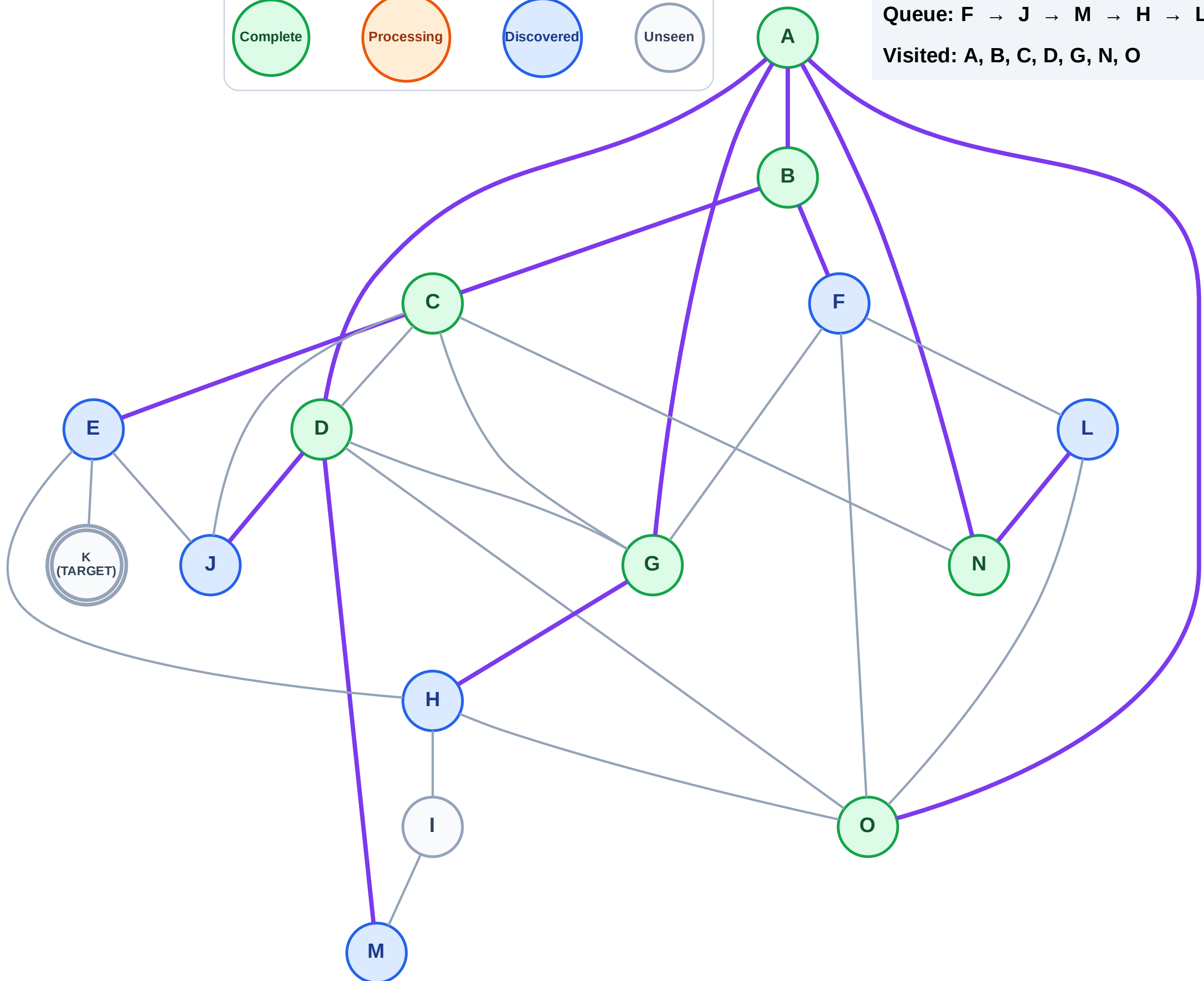
Discovered



Unseen

Queue: F → J → M → H → L → E

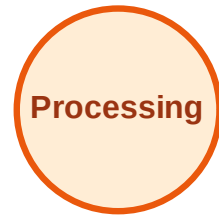
Visited: A, B, C, D, G, N, O



BFS Traversal

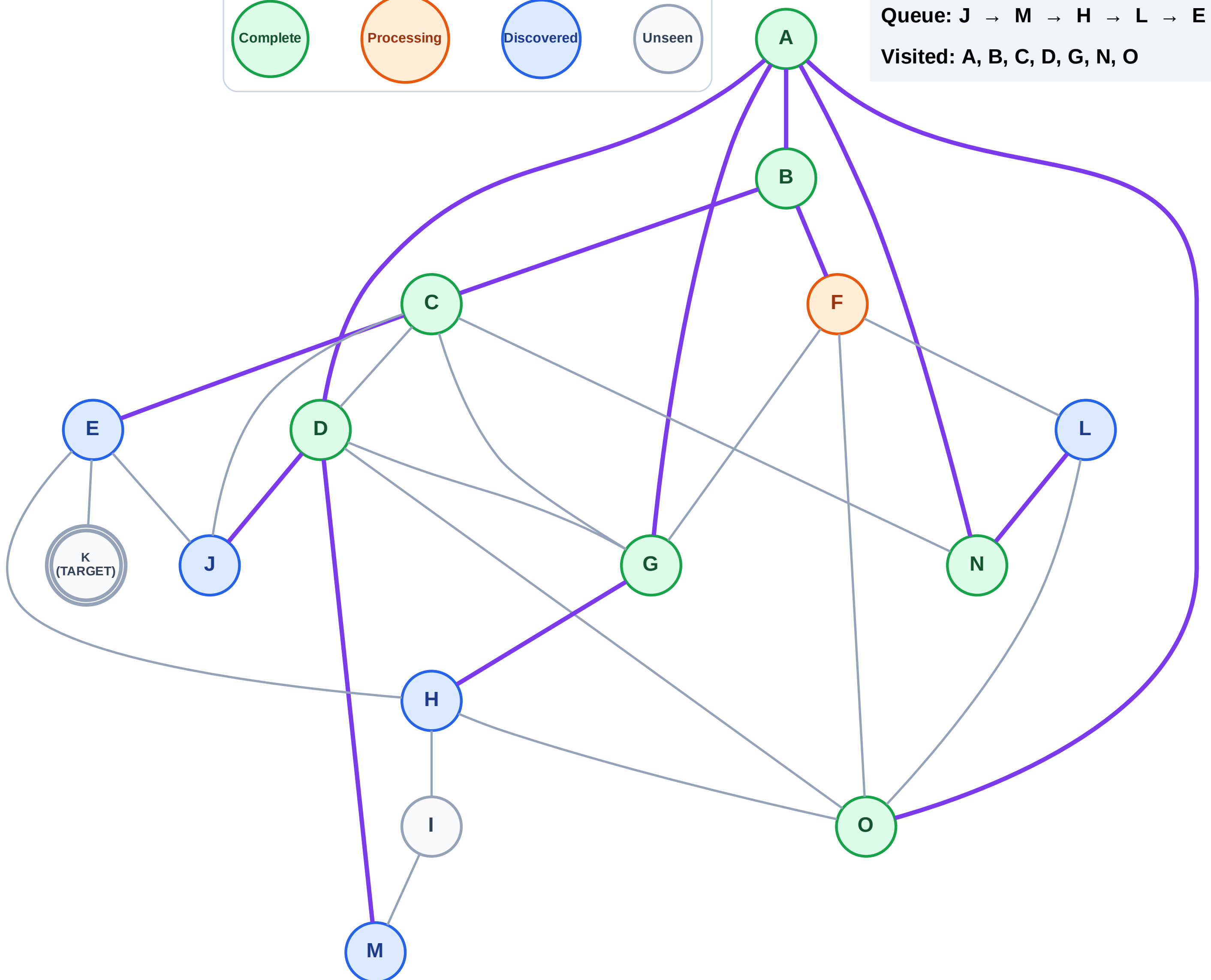
Step 16: Process node F

Legend



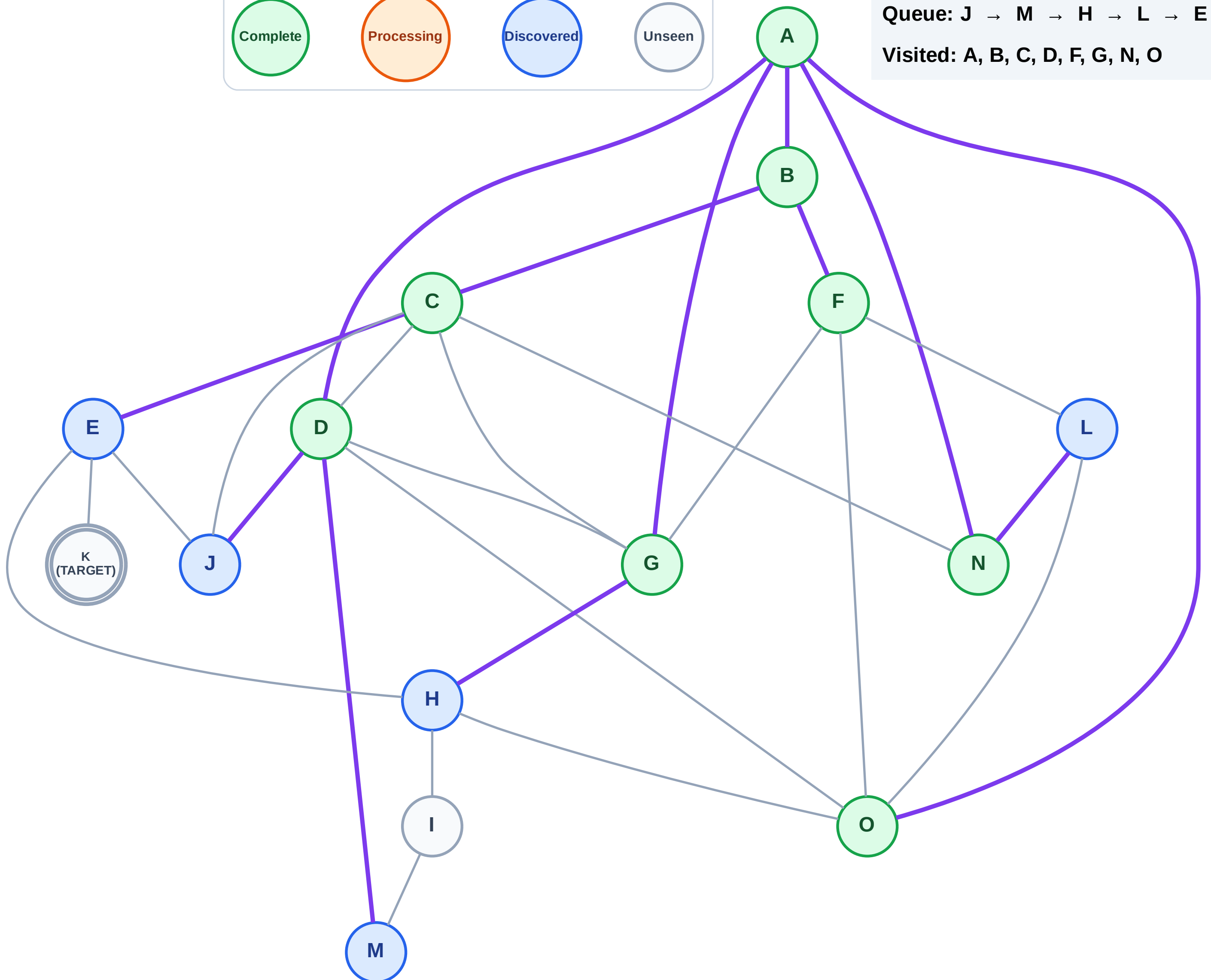
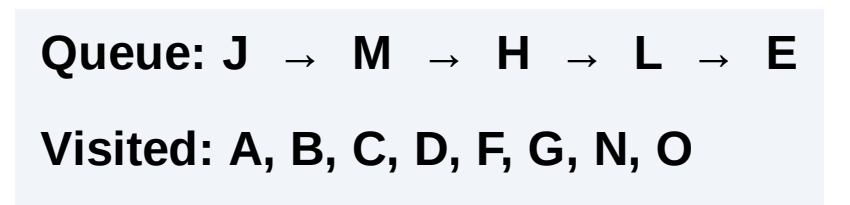
Queue: J → M → H → L → E

Visited: A, B, C, D, G, N, O



Step 17: No new neighbors from F

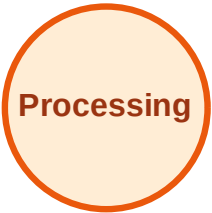
Step 17: No new neighbors from F



BFS Traversal

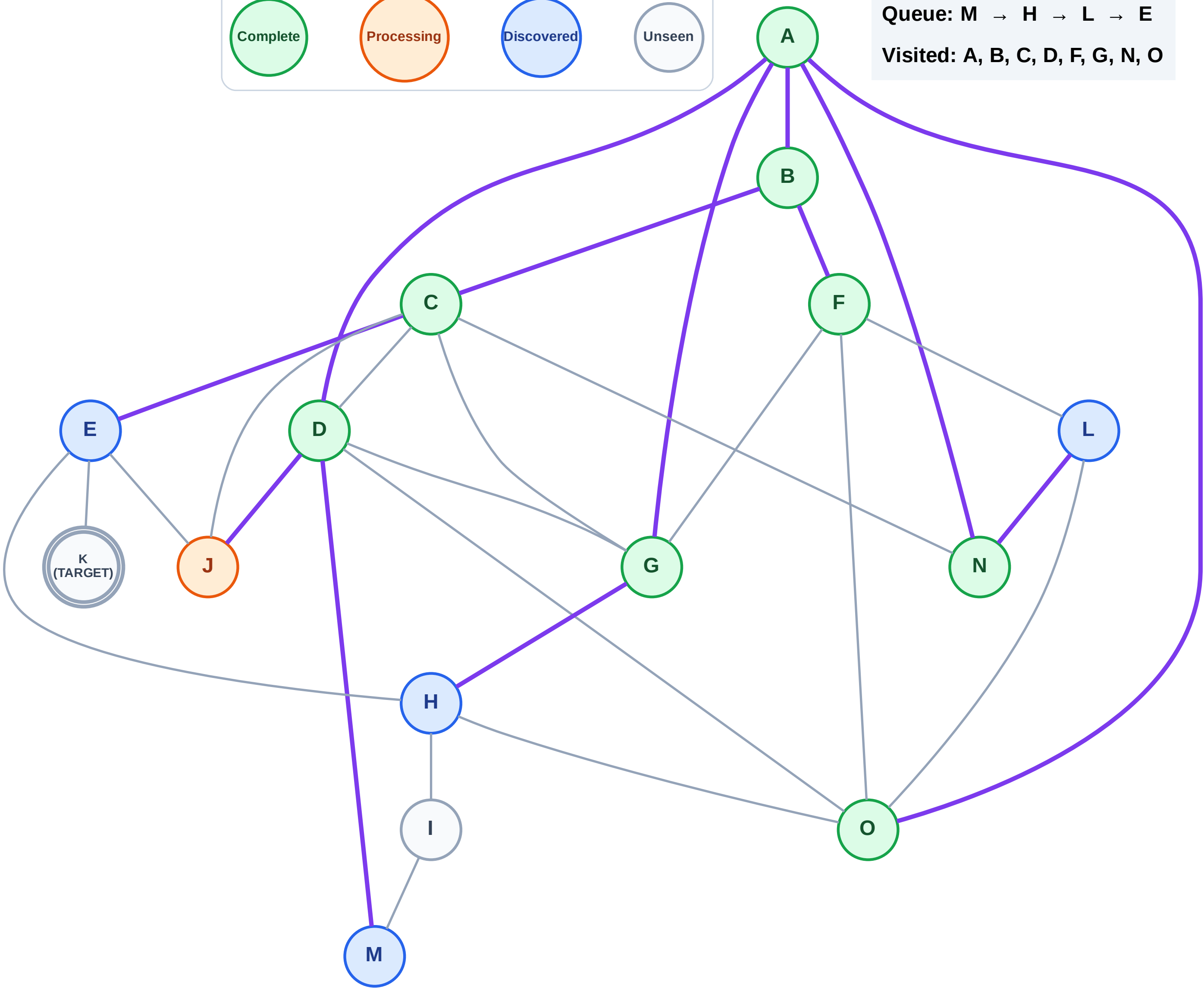
Step 18: Process node J

Legend



Queue: M → H → L → E

Visited: A, B, C, D, F, G, N, O



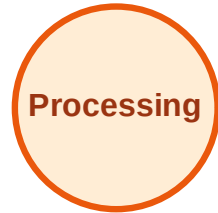
BFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



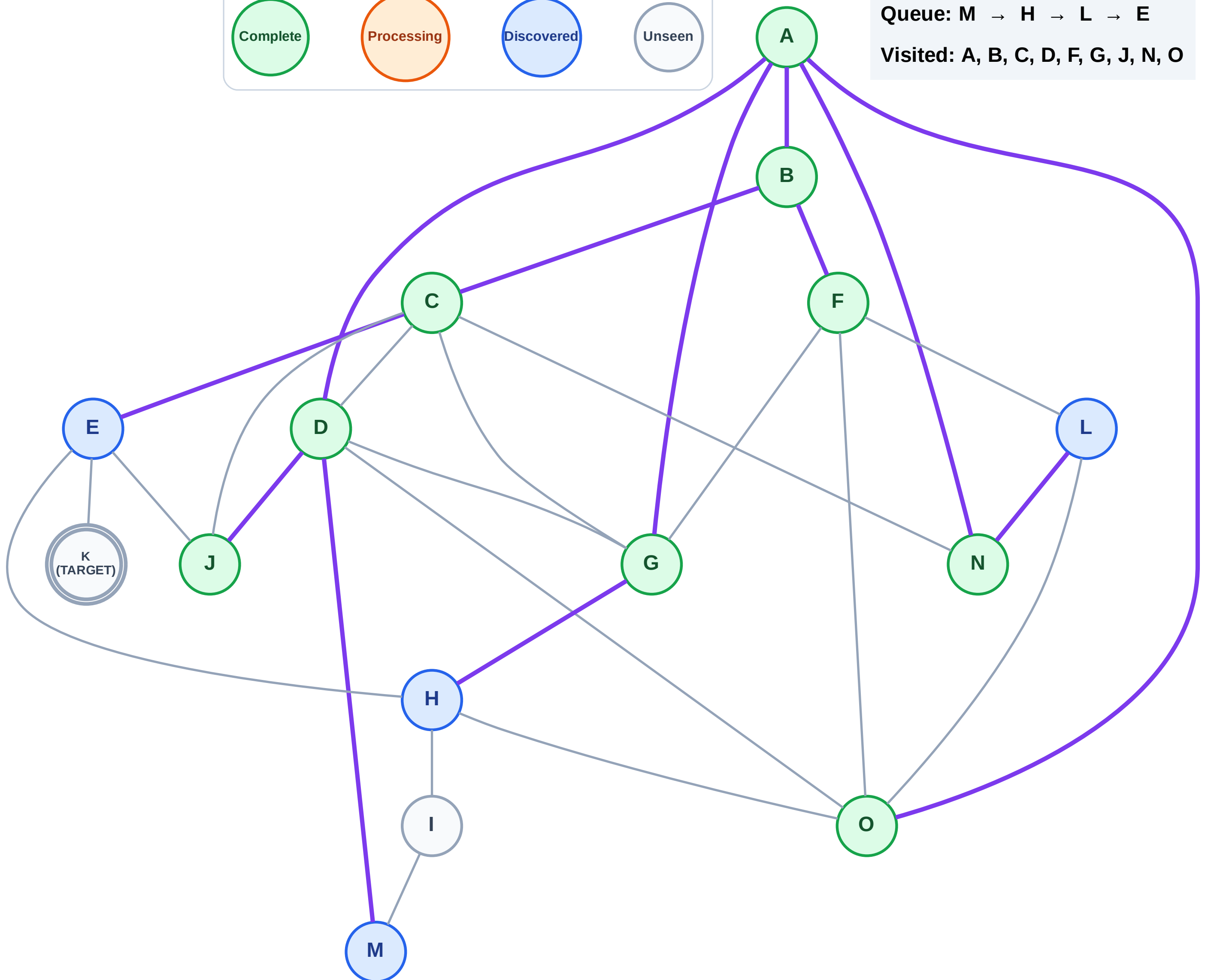
Discovered



Unseen

Queue: M → H → L → E

Visited: A, B, C, D, F, G, J, N, O



BFS Traversal

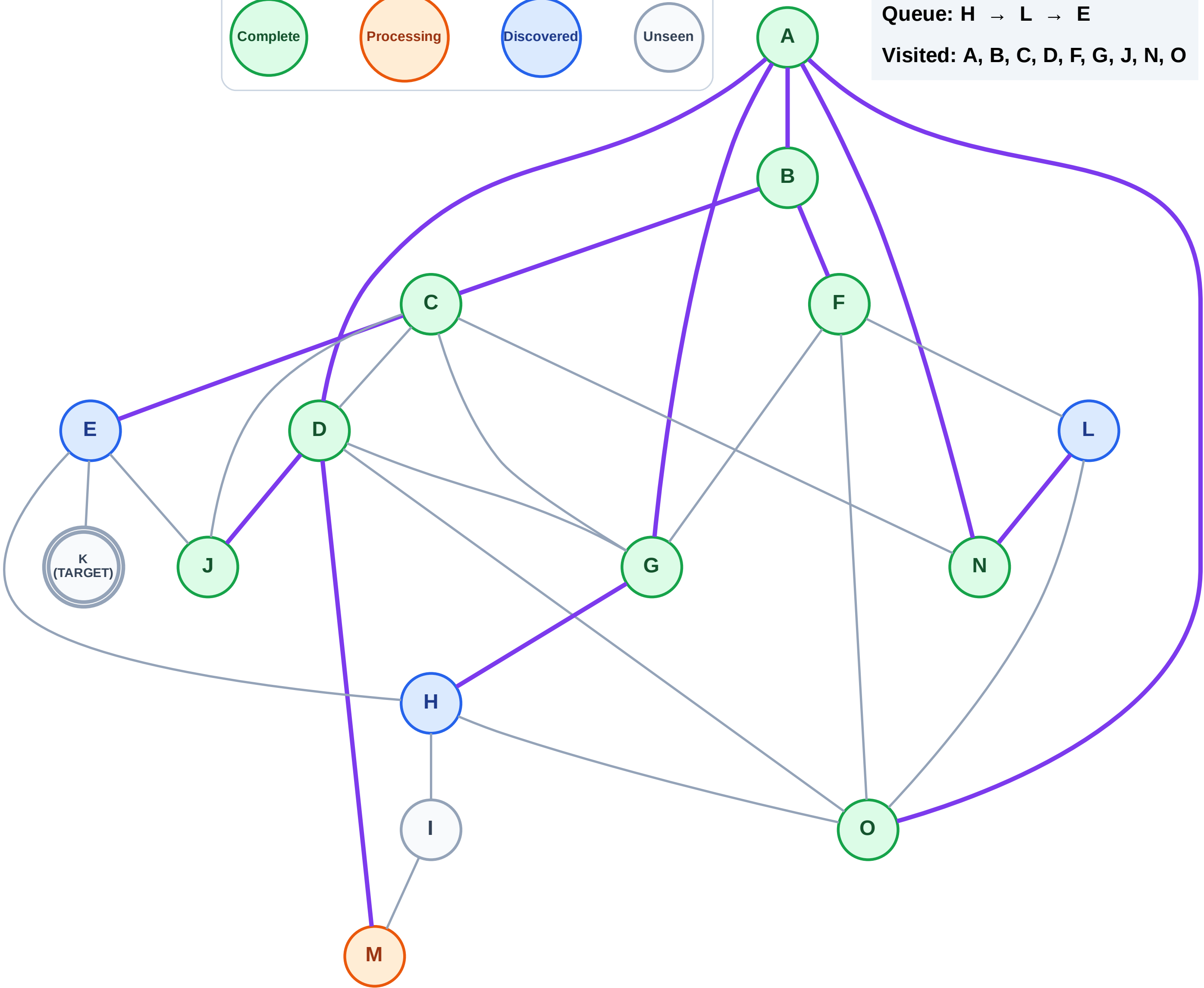
Step 20: Process node M

Legend



Queue: H → L → E

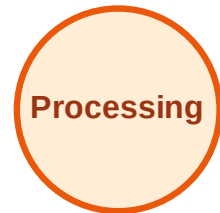
Visited: A, B, C, D, F, G, J, N, O



BFS Traversal

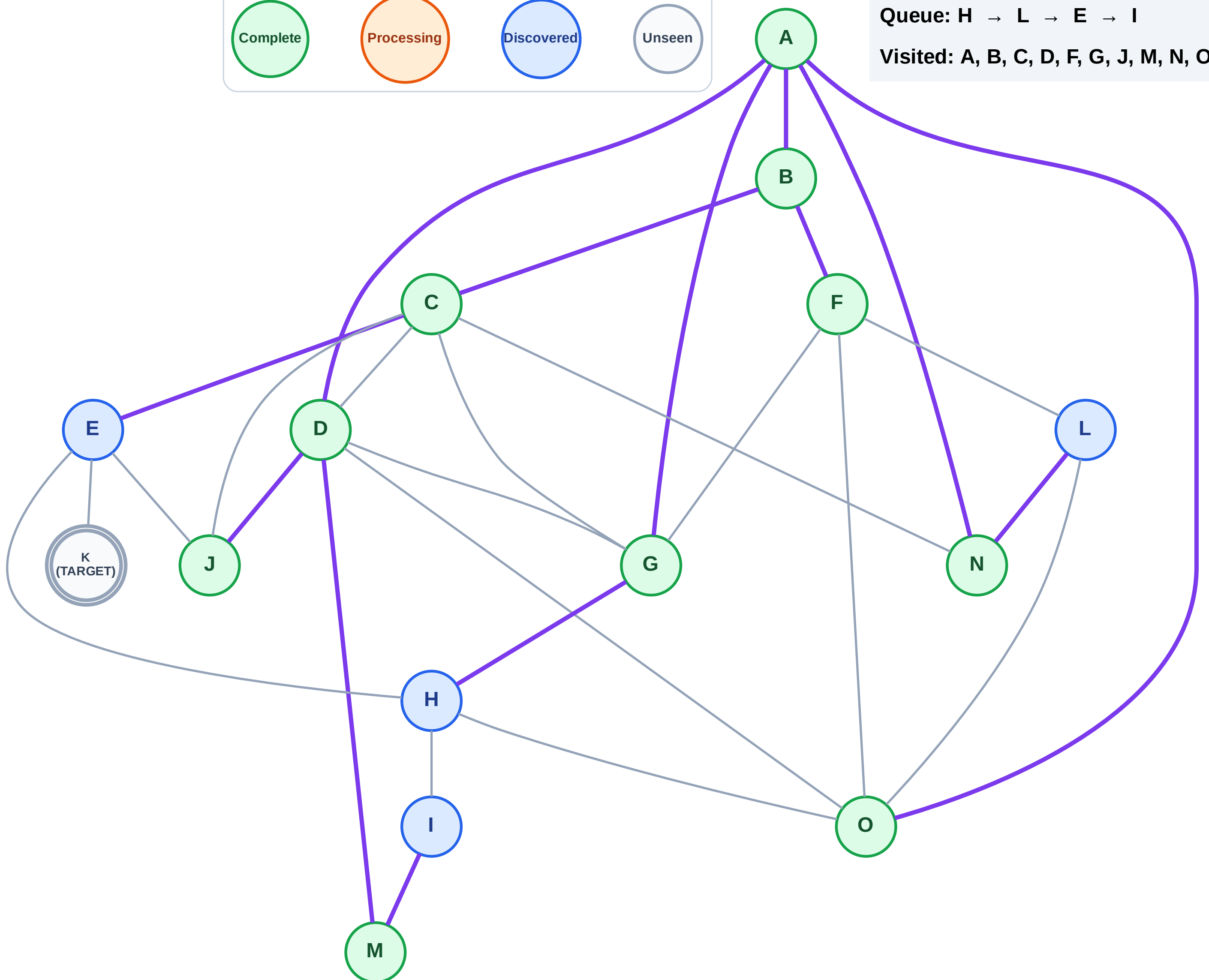
Step 21: Discover I from M

Legend



Queue: H → L → E → I

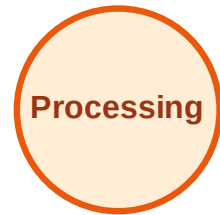
Visited: A, B, C, D, F, G, J, M, N, O



BFS Traversal

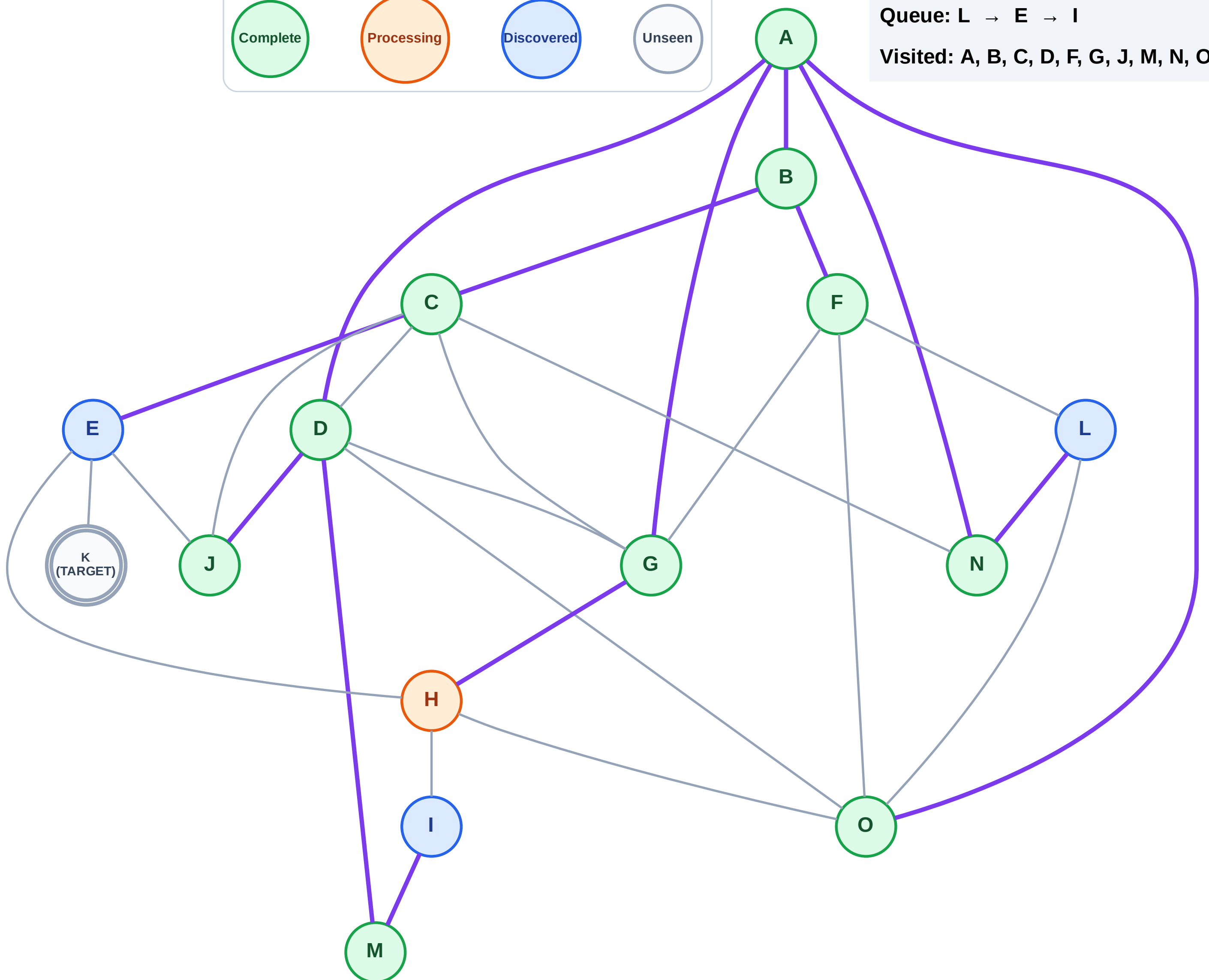
Step 22: Process node H

Legend



Queue: L → E → I

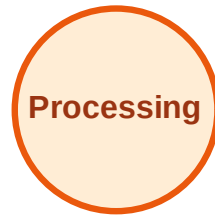
Visited: A, B, C, D, F, G, J, M, N, O



BFS Traversal

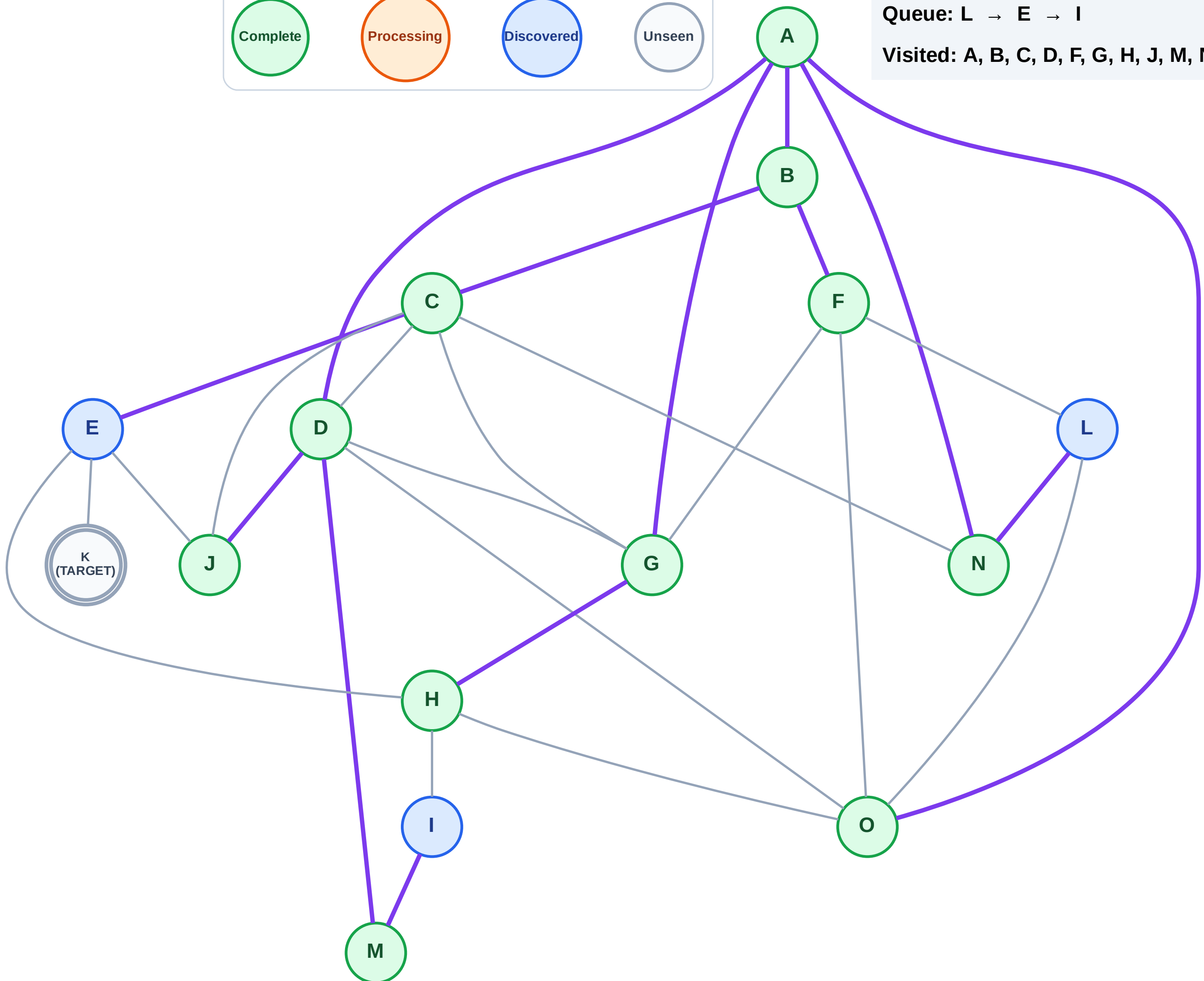
Step 23: No new neighbors from H

Legend



Queue: L → E → I

Visited: A, B, C, D, F, G, H, J, M, N, O



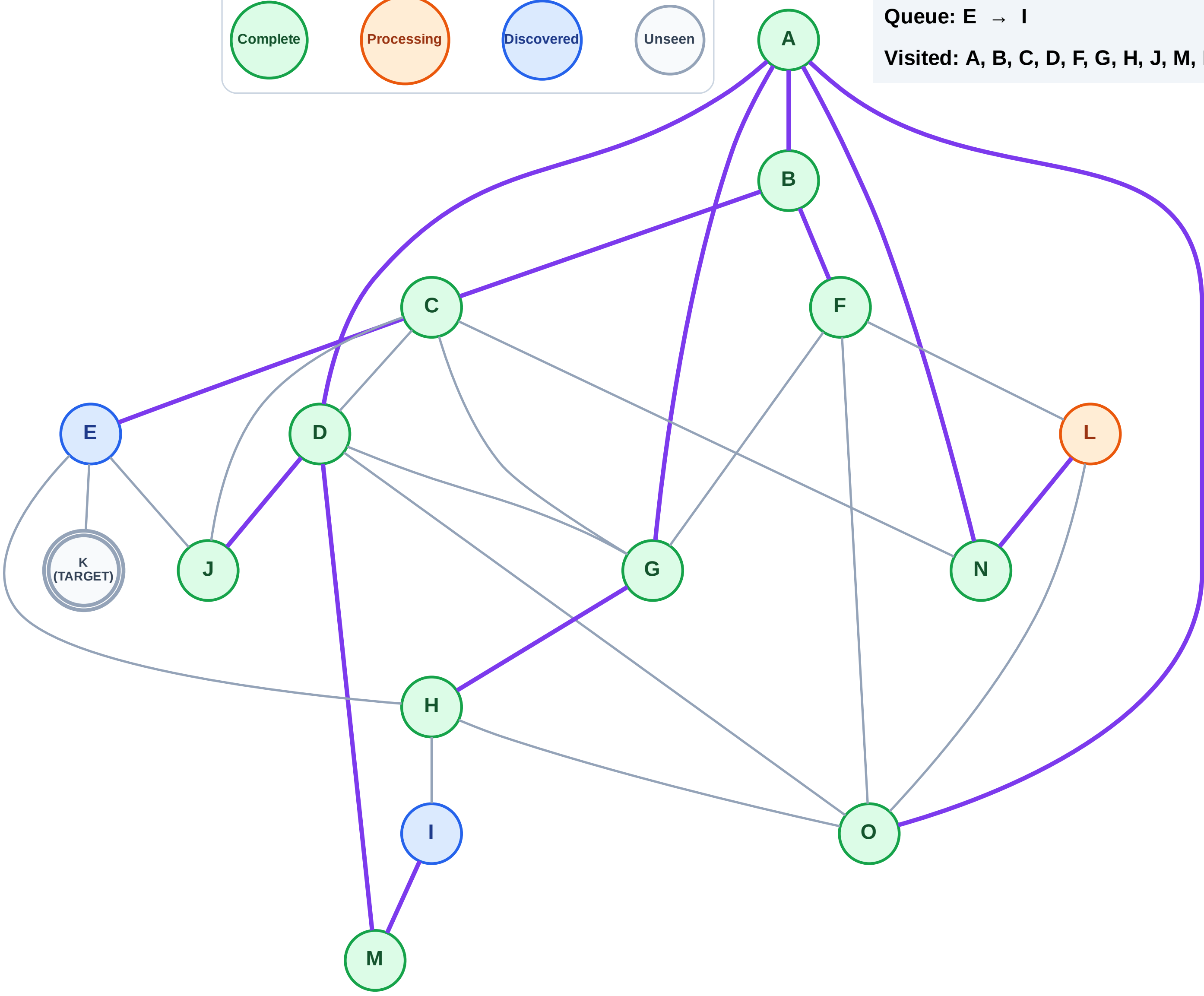
BFS Traversal

Step 24: Process node L

Legend



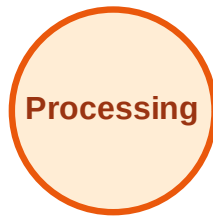
Queue: E → I
Visited: A, B, C, D, F, G, H, J, M, N, O



BFS Traversal

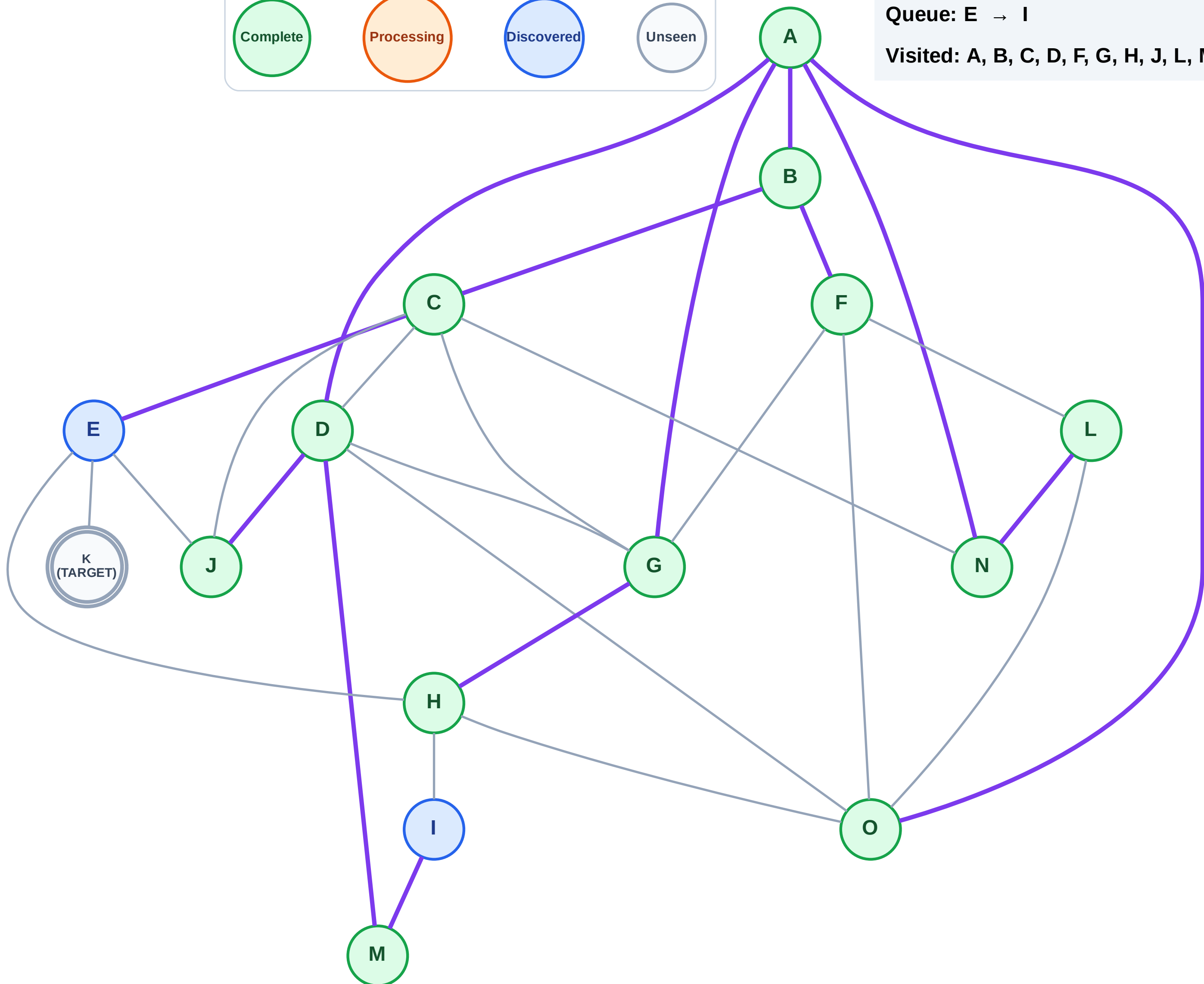
Step 25: No new neighbors from L

Legend



Queue: E → I

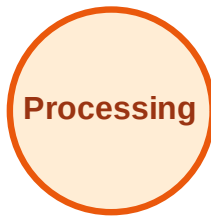
Visited: A, B, C, D, F, G, H, J, L, M, N, O



BFS Traversal

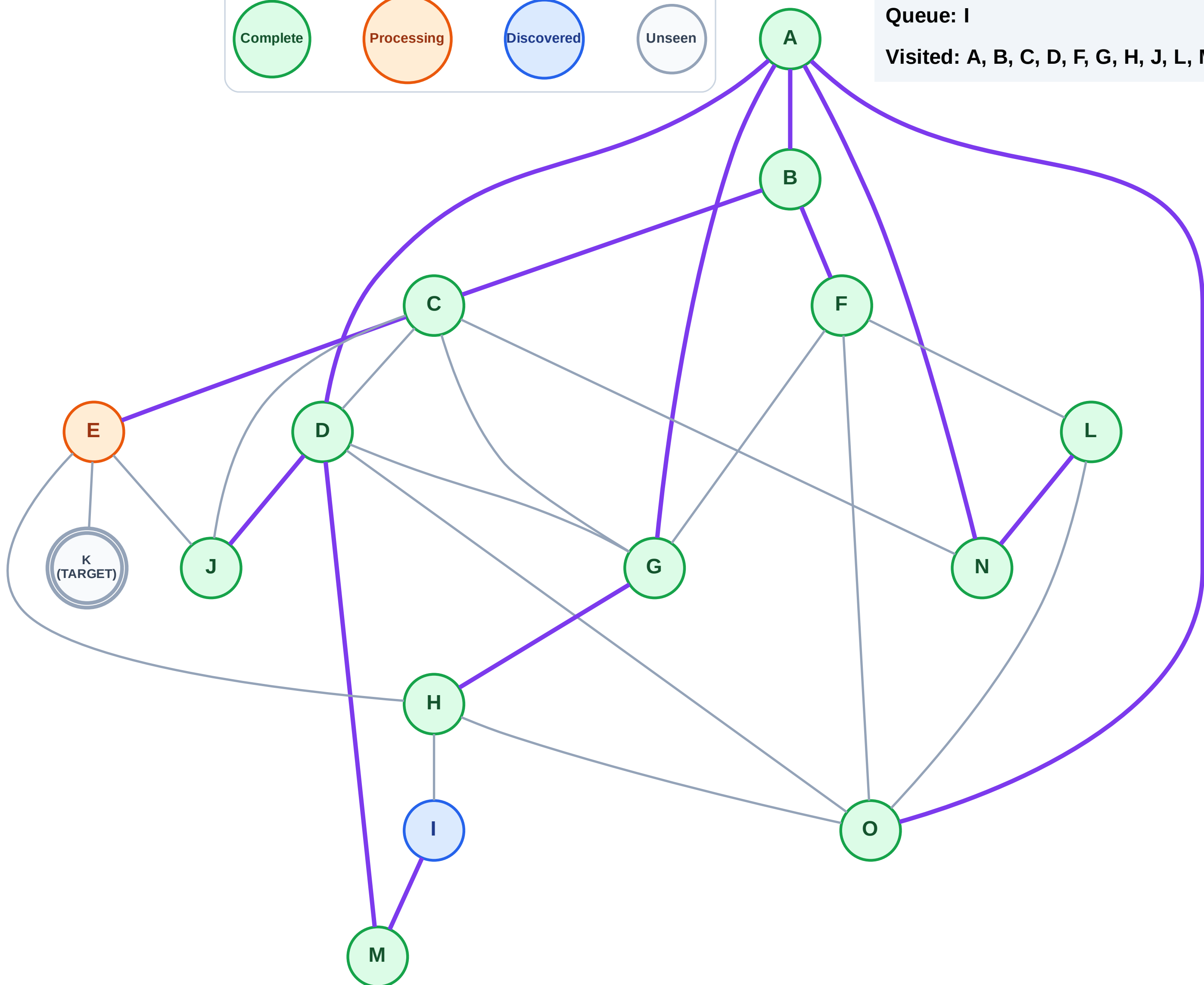
Step 26: Process node E

Legend



Queue: I

Visited: A, B, C, D, F, G, H, J, L, M, N, O



BFS Traversal

Step 27: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → E → K

Legend

Complete

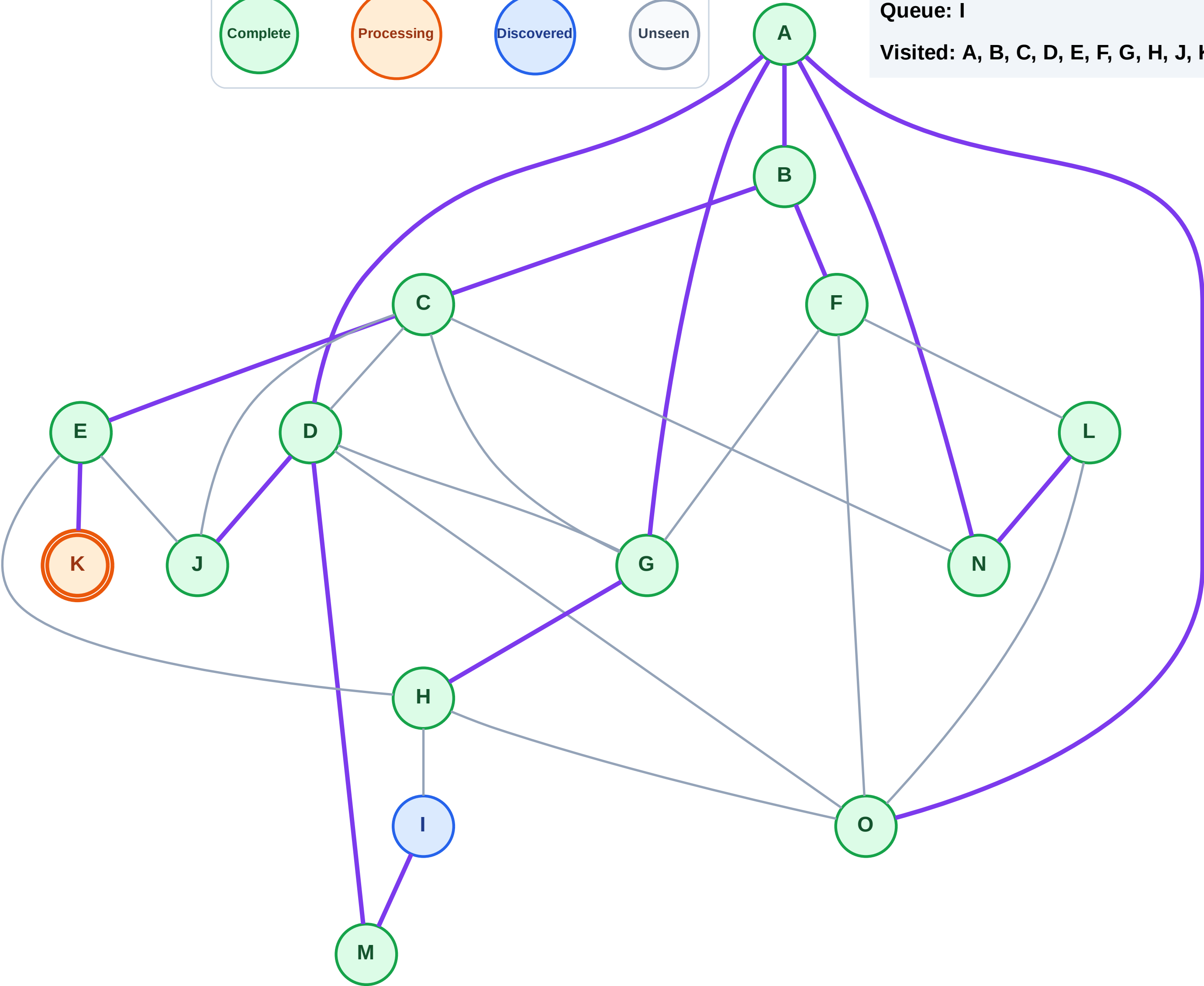
Processing

Discovered

Unseen

Queue: I

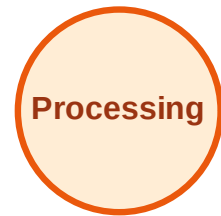
Visited: A, B, C, D, E, F, G, H, J, K, L, M, N, O



DFS Traversal

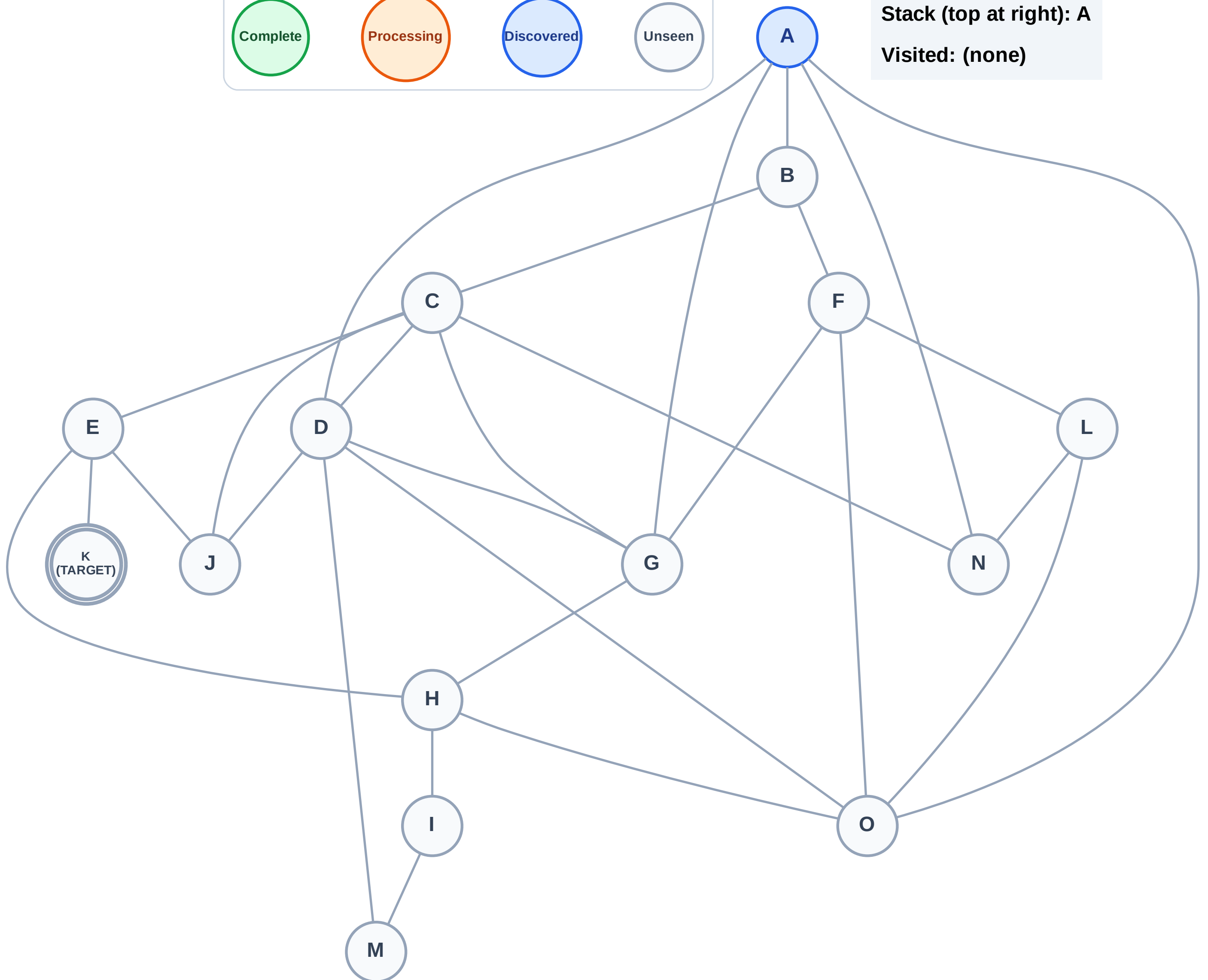
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

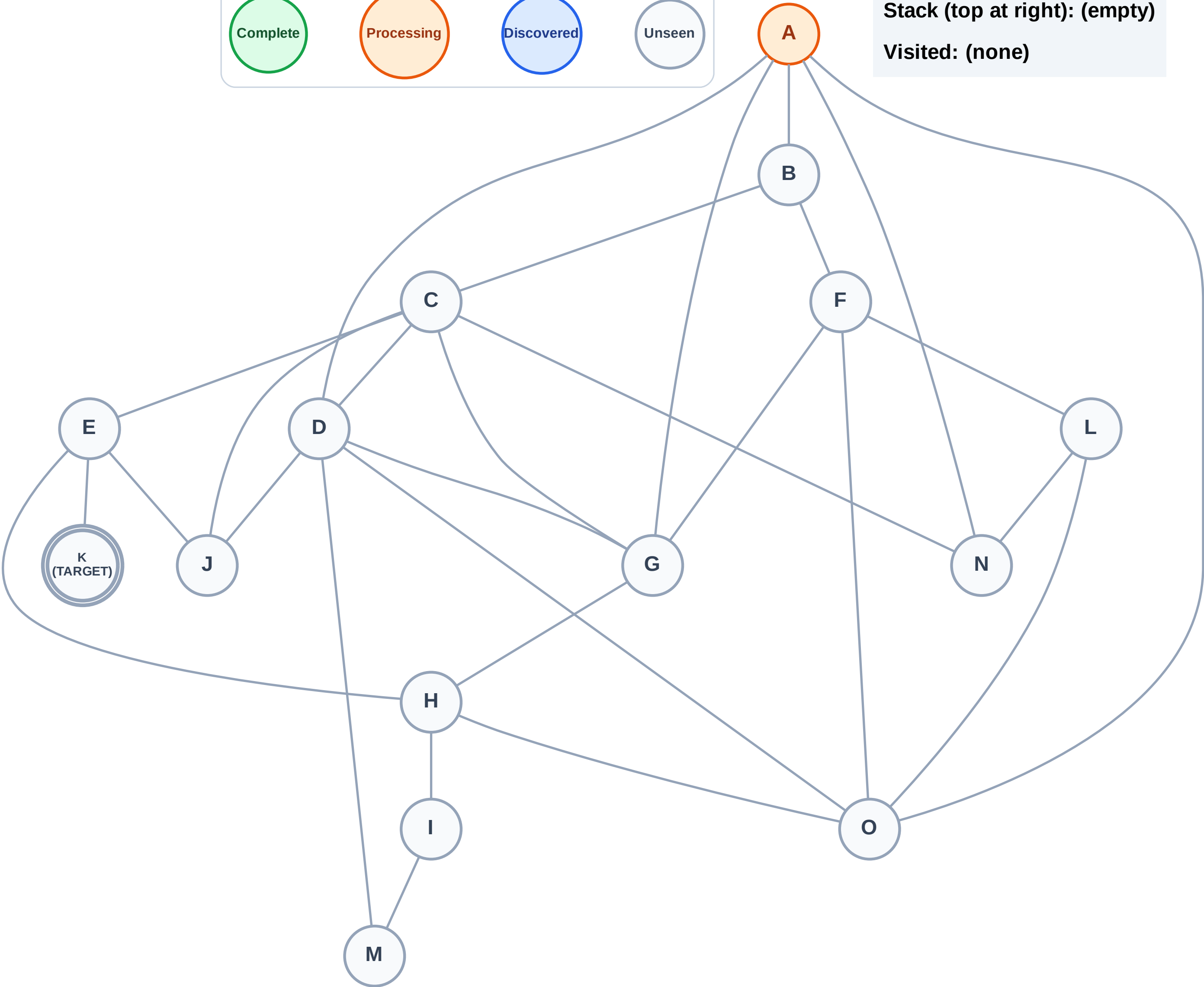
Complete

Processing

Discovered

Unseen

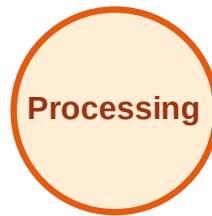
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

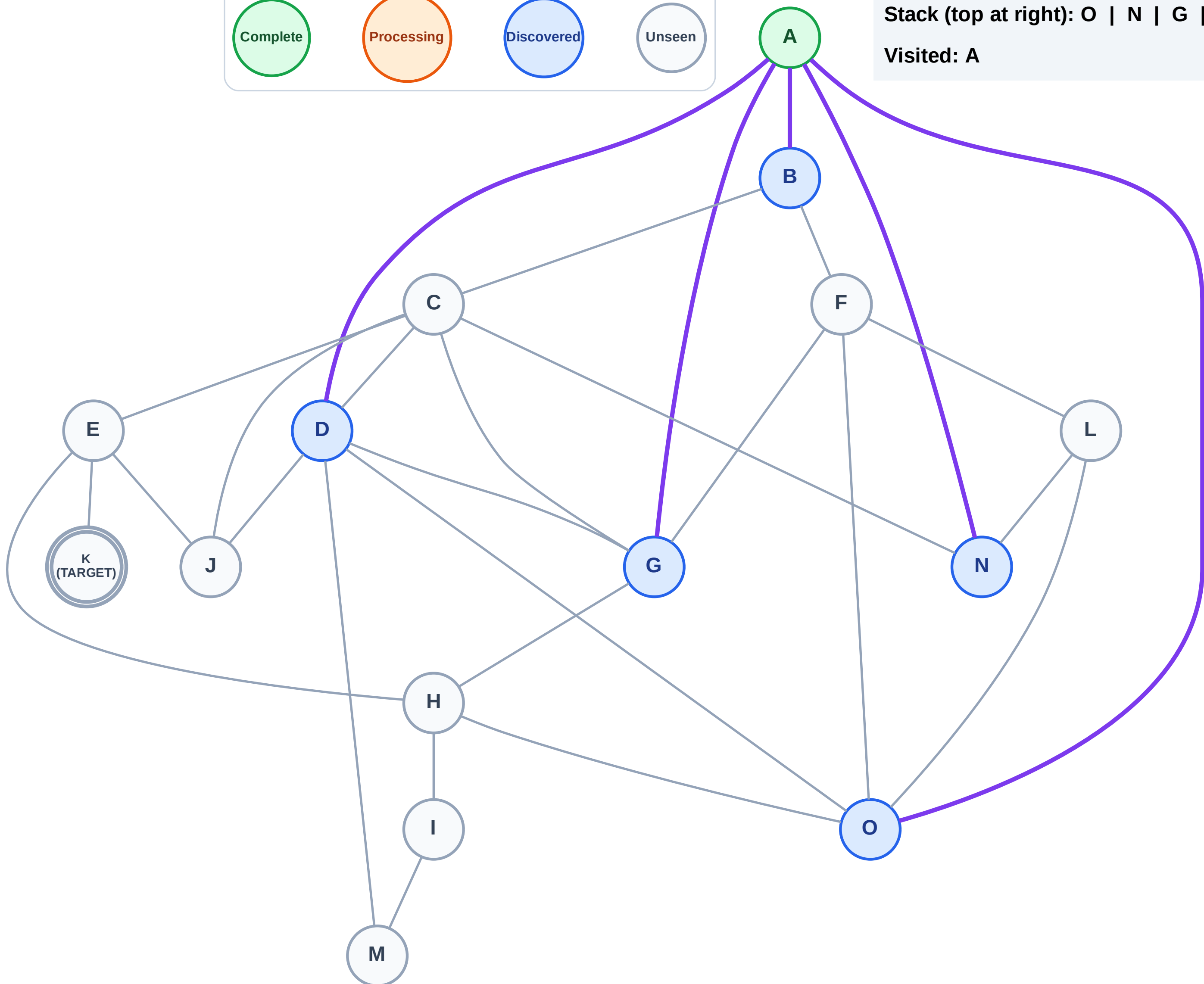
Step 3: Discover O, N, G, D, B from A

Legend



Stack (top at right): O | N | G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

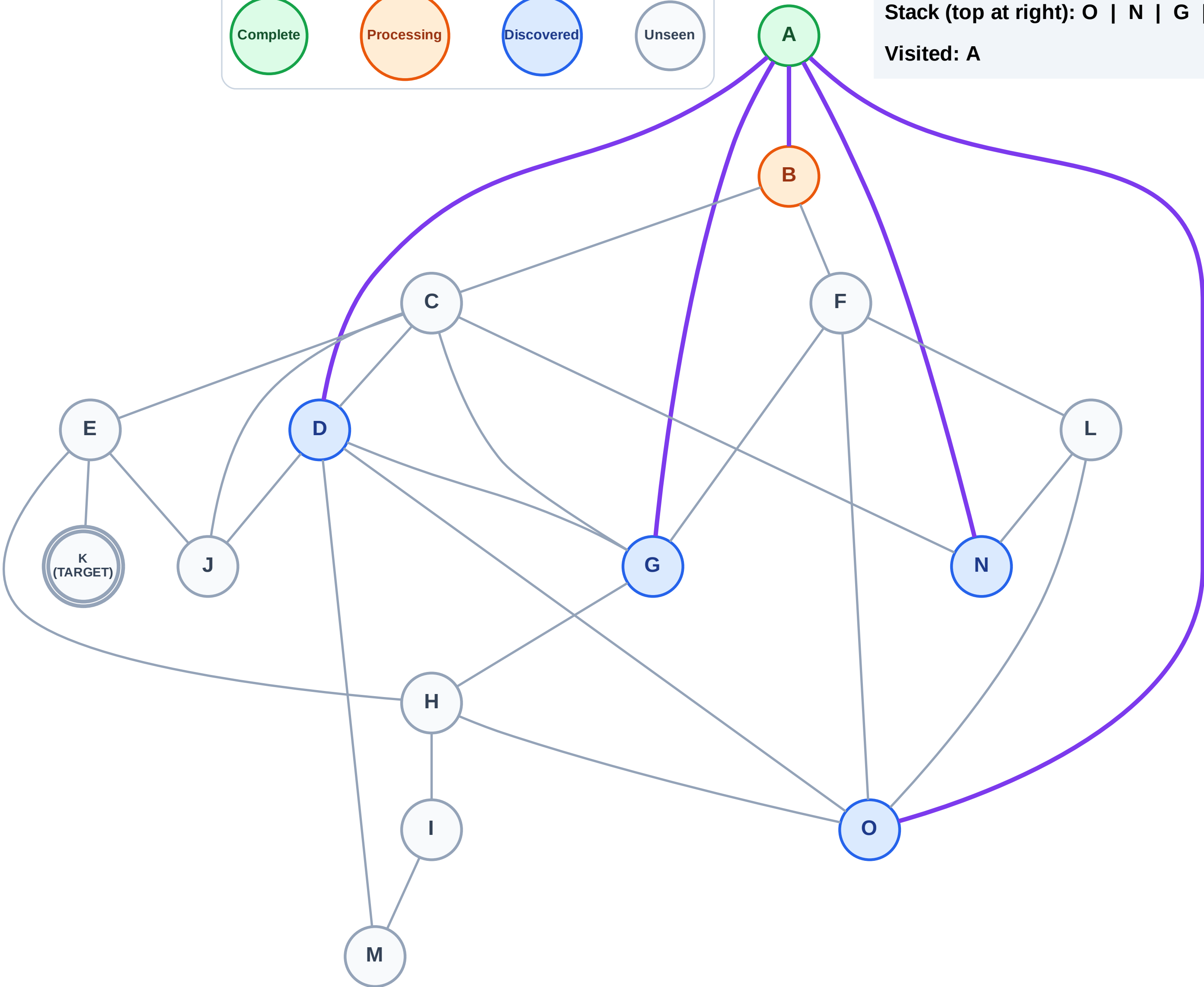
Processing

Discovered

Unseen

Stack (top at right): O | N | G | D

Visited: A



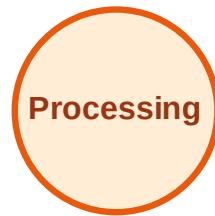
DFS Traversal

Step 5: Discover F, C from B

Legend



Complete



Processing



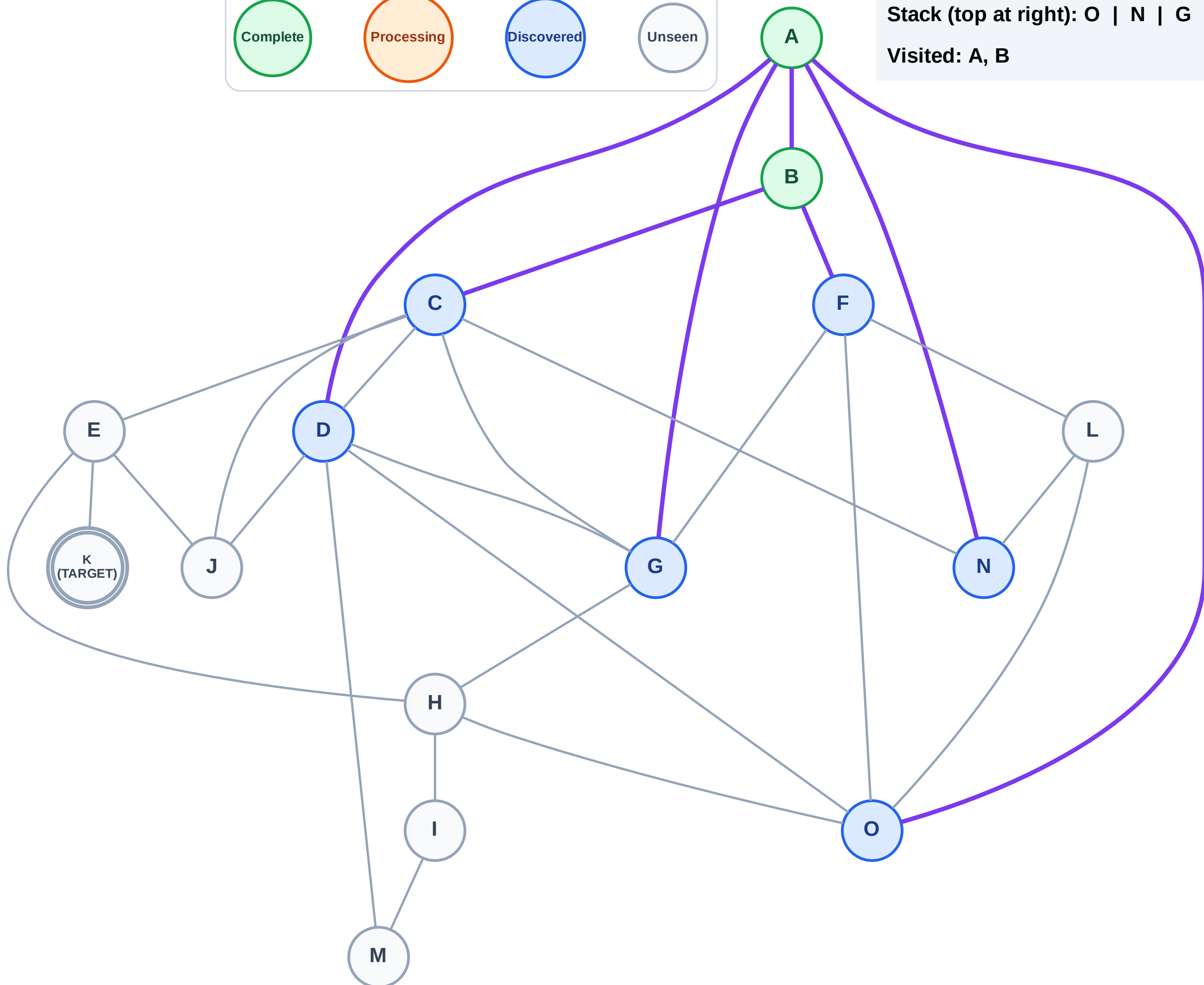
Discovered



Unseen

Stack (top at right): O | N | G | D | F | C

Visited: A, B



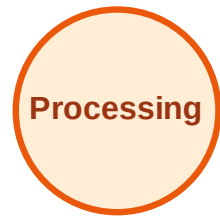
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



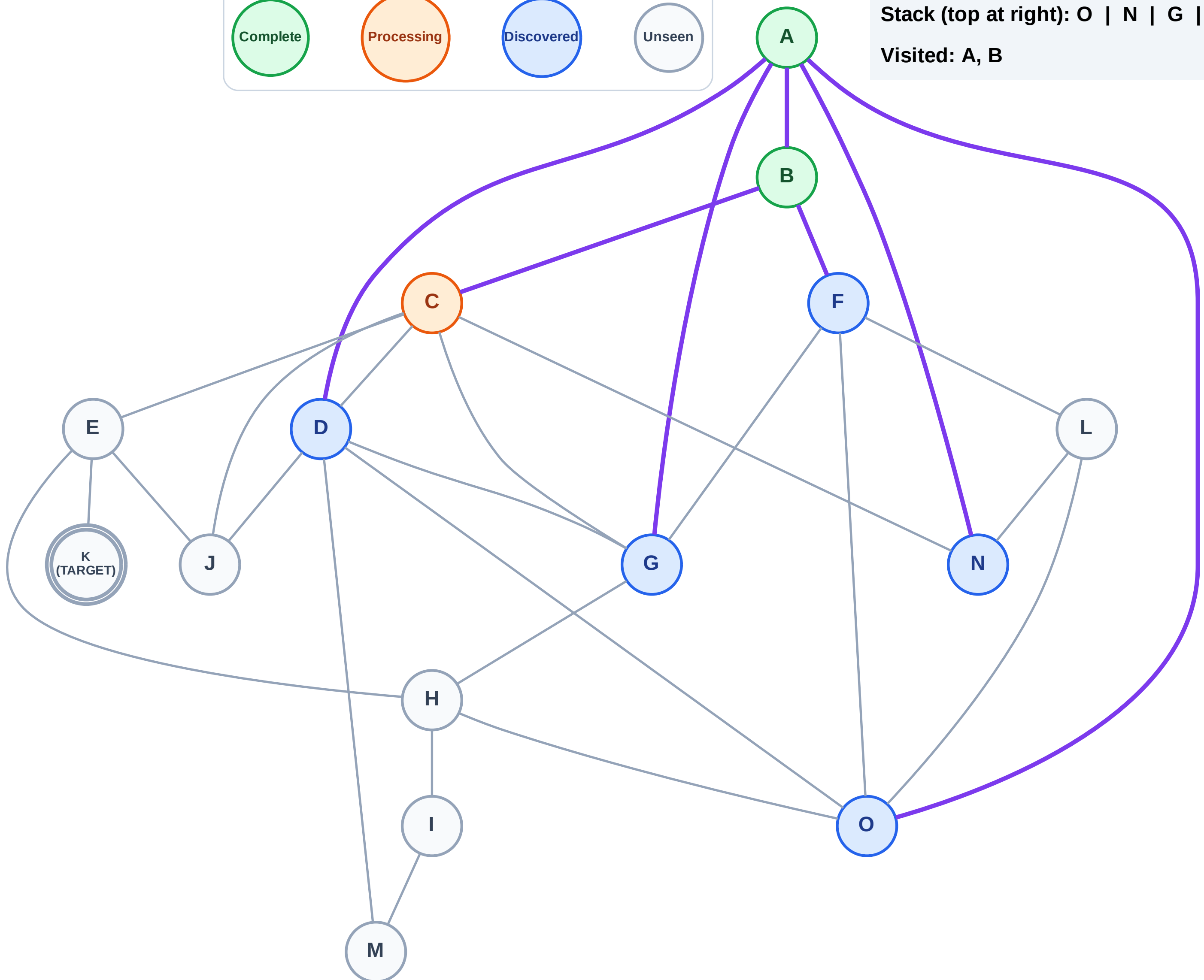
Discovered



Unseen

Stack (top at right): O | N | G | D | F

Visited: A, B



Step 7: Discover J, E from C

Step 7: Discover J, E from C

Complete

Complete



Processing

Processing



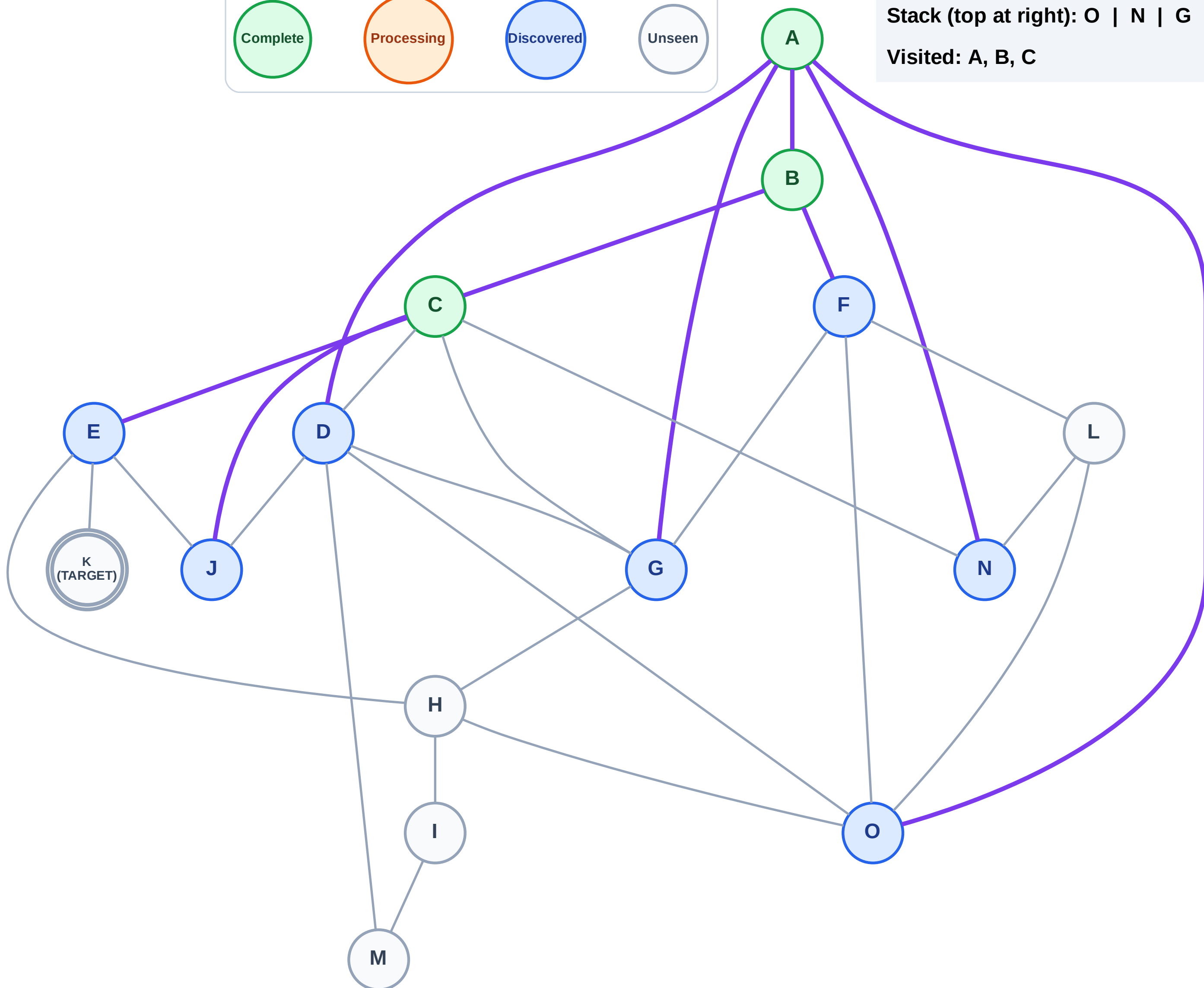
Discovered

Discovered



Unseen

Visited: A, B, C



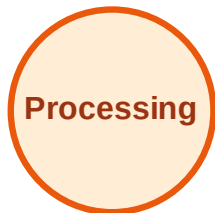
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



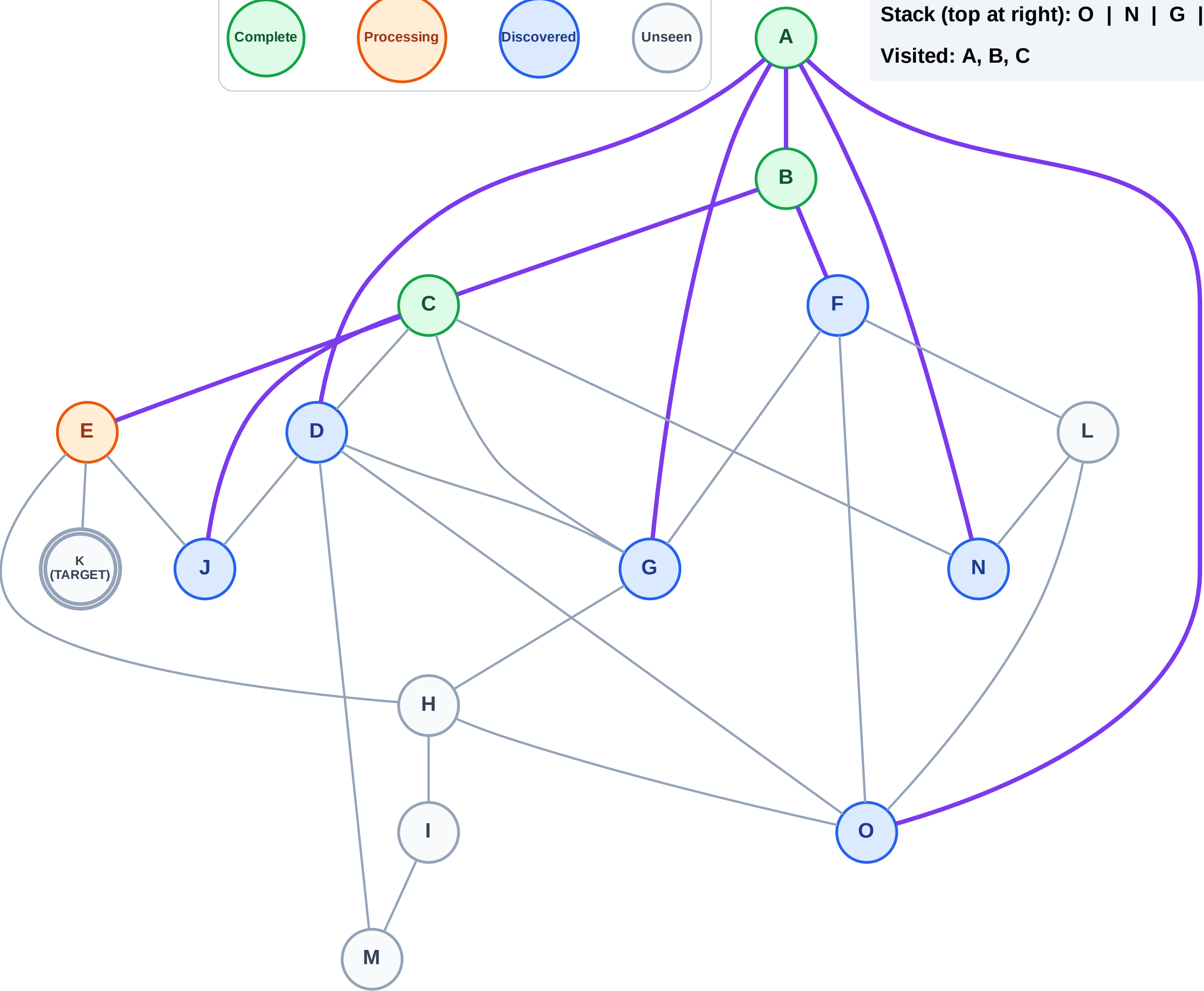
Discovered



Unseen

Stack (top at right): O | N | G | D | F | J

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | G | D | F | J

Visited: A, B, C, E, K

